# Point-by-Point Response to Reviewer #1 & Manuscript Revision Guide

**Manuscript:** *Structural Basis for MCAM Mediated Inhibition of Laminin Dependent Cell Migration* (Li, Alarcon-Frias, Li, Alarcón, & Klein; `PMC13596217`)

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## Executive Strategy: How Our New Quantitative Biophysics & Model Directly Answer Reviewer #1

Reviewer #1 praised the clarity of the writing, the novelty of the cryo-EM structures, and the experimental execution, but raised **two connected concerns**:

1. **Biological / Physiological Story:** *Why does the MCAM–Laminin-411/421 interaction exist, and how does it fit into physiological cell migration if inhibition was only shown by adding soluble MCAM ( `sMCAM` )?*
2. **Affinity & Therapeutic Framing:** *Why did soluble MCAM require 1 μM in Transwell assays ("very low affinity"), and why propose `sMCAM` as a therapeutic lead when an anti-Laminin $\alpha_4$ antibody could be used?*

**Why the new MIXA quantitative biophysical analysis + 2D/3D model is the exact missing piece:**

In the original manuscript, the kinetics were approximated by a single 1:1 fit ($K_D \approx 41.5$ nM) and the $\alpha_6\beta_1$ integrin competition was qualitative. By adding our **quantitative two-site kinetic dissection** and **1:1 mass-action solution competition titration (N = 3 replicates)**, we uncover the **exact quantitative rules** that govern this system:

1. **MCAM is NOT a "very low affinity" micromolar binder:** Global kinetic analysis (N = 3, $R^2 = 0.997$) resolves two distinct bound states on immobilized MCAM-Fc: **$K_{D,1}$ = 4.63 ± 0.48 nM (Combined High-Affinity State — Both Sites Engaged: $\beta_1\gamma_1$ coiled-coil + $\alpha_4$ LG1–LG3, $k_{off,1} = 2.53 \times 10^{-3}$ $s^{-1}$, $t_{1/2}$ = 278 s)** and **$K_{D,2}$ = 34.4 ± 2.3 nM (Low-Affinity State — 1 Site Bound / Partially Engaged, $k_{off,2} = 3.88 \times 10^{-2}$ $s^{-1}$, $t_{1/2}$ = 18.0 s)** (mathematically parameterized as apparent stepwise constants $K_{D,step\ 1}$, $K_{D,step\ 2}$). Crucially, because MCAM-Fc is immobilized on the Protein A biosensor surface via its Fc anchor (no solution Fc avidity) and removing the $\beta_1\gamma_1$ chains in isolated `LamA4G123` on the **exact same MCAM-Fc surface** collapses 86.1% of binding to a fast-off 1-site state ($K_{D,2}$ = 824 ± 36 nM, $t_{1/2}$ = 6.2 s), **$K_{D,1}$ = 4.63 nM is definitively the combined $K_D$ when both sites on intact Laminin-411 are engaged**.

2. **However, $\alpha_6\beta_1$ integrin is >6× tighter ($K_D$ = 0.73 ± 0.22 nM):** Quantitative solution equilibrium titration (N = 3, $R^2 = 0.982$) demonstrates that $\alpha_6\beta_1$ integrin binds the overlapping `LG1–LG2` site with **sub-nanomolar affinity (0.73 ± 0.22 nM under linear response mapping; 0.26 ± 0.08 nM conditional on a saturable surface-capture detector model)**—more than **6-fold tighter** than MCAM's combined both-sites-engaged $K_{D,1}$ (4.63 nM) and **~ 47-fold tighter** than the 1-site bound $K_{D,2}$ (34.4 nM).

3. **This >6× affinity gap + 2D spatial restriction resolves the central physiological paradox of MCAM in the literature:**

- **In 2D (membrane-anchored MCAM):** When MCAM and $\alpha_6\beta_1$ integrin are co-expressed on the cell surface, both MCAM and matrix Laminin-411 are **spatially restricted in 2D**. Because $\alpha_6\beta_1$ integrin has >6× higher

affinity, membrane-tethered MCAM cannot outcompete integrin or globally deplete the matrix—so **$\alpha_6\beta_1$ integrin wins locally and drives cell migration**.

- **In 3D (proteolytically shed soluble `sMCAM` / `sCD146`):** In vascular and tumor microenvironments, membrane MCAM is physiologically cleaved by metalloproteinases (e.g., MT1-MMP/MMP-14) to release **soluble `sMCAM` (`sCD146`)**. Freed from 2D spatial restriction, high local pericellular `sMCAM` diffuses in 3D to pre-occupy Laminin-411 `LG1–LG2` across the matrix, acting as a **proteolytic switch / negative-feedback brake** that sterically blocks $\alpha_6\beta_1$ integrin engagement and halts migration.

- **Why 1 µM `sMCAM` is required in Transwell assays:** By Cheng–Prusoff mass action, for a 4.6–34 nM soluble decoy to competitively displace a **0.72 nM integrin receptor** that is further enhanced by multivalent 2D cell-surface avidity, the soluble competitor must be supplied in excess (0.2–1 µM) to maintain >95% fractional saturation of matrix `LG1–LG2`.

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## Draft Point-by-Point Response to Reviewer #1

### Reviewer Comment 1.1 — Summary & Cryo-EM Resolution

> *"In this paper, Li et al. determined the 3D structure of complex between fragments of laminin 411/421 and MCAM ectodomain by cryo-EM. The interface overlapped with predicted integrin-binding region within laminin α4 LG1-3, which was validated by the inhibition of MCAM binding by preincubating LN411 with α6β1 integrin. The resolution and the quality of the final models are at the lower end of present-day average for cryo-EM structures, which is understandable considering the highly mobile nature of the multi-modular extracellular proteins. The authors validated the authenticity of the MCAM-LN411 interface by performing binding assays using limited set of single-residue mutants. Furthermore, the authors showed that addition of soluble MCAM fragment attenuated integrin-dependent cell migration in a transwell assay, proposing that this may have therapeutic potential to block integrin-mediated leukocytes or tumor cell trafficking through LN α4-containing vascular basement membrane."*

**Response:**

We thank Reviewer #1 for the thoughtful assessment of our cryo-EM structures and experimental validation, and for recognizing the technical challenge of resolving dynamic, multi-modular extracellular matrix–receptor complexes (`MCAM` bound to `Laminin-411/421`). Notably, our structures and mutagenesis define that the shared receptor footprint is restricted specifically to the **`LG1–LG2` module** of Laminin $\alpha_4$ (interacting with MCAM `V2–C1`), whereas **`LG3` is uninvolved**.

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### Reviewer Comment 1.2 — Physiological Significance & Mechanistic Integration into Cell Migration

> *"The (patho)physiological importance of the MCAM-LN411/421 interaction, either on its own or in the context of modulation of integrin-mediated adhesion, remains highly obscure, making the biological impact of mere structure determination of the complex low. It would have been much more exciting if the study explores in vivo function of MCAM in leukocyte trafficking or tumor dissemination by using the structural information obtained here. As the authors themselves admit, these are experimental results obtained using purified proteins only. If MCAM has potential to modulate cell migration behavior in the physiological settings, then the interaction has to be evaluated in physiologically relevant systems, ideally in animals. Showing the presence of specific molecular interaction is one thing, but explaining how that interaction fits into the complex molecular events that are orchestrated into particular cell/tissue behavior is another."*

**Response:**

We completely agree with the Reviewer that showing a structural interaction is only the first step, and that explaining **how the MCAM–Laminin-411 interaction fits into the quantitative molecular events governing cell migration** is essential for the broad readership of *Nature Communications*.

Prompted by this comment, we performed a rigorous, multi-replicate quantitative biophysical dissection of both **direct MCAM–Laminin-411 binding kinetics** (N = 3 independent BLI replicates) and **quantitative solution equilibrium competition between $\alpha_6\beta_1$ integrin and MCAM for Laminin-411** (N = 3 independent BLI replicates), and integrated these findings into a new **mechanistic conclusion model (new Figure / Schematic)** that resolves a long-standing paradox in the MCAM and laminin fields:

1. **The Paradox in the Field:**

Previous studies established Laminin-411 ($\alpha_4\beta_1\gamma_1$) as a physiological vascular basement membrane ligand for both **MCAM** (driving TH17 lymphocyte and melanoma adhesion; Flanagan et al., *PLoS ONE* 2012; Ishikawa et al., *PNAS* 2014) and **$\alpha_6\beta_1$ integrin** (driving cell migration). However, it remained unexplained why **membrane-expressed MCAM** on melanoma or endothelial cells does not block $\alpha_6\beta_1$ integrin–driven migration on Laminin-411, whereas **soluble MCAM (`sMCAM` / `sCD146`)**—which is physiologically shed from cells by matrix metalloproteinases (such as MT1-MMP/MMP-14)—potently inhibits migration.

2. **Quantitative Resolution via Two-Site Surface-Capture Kinetics & Solution Competition:**

   • **Combined Both-Sites-Engaged (K_D,1) vs. 1-Site-Bound (K_D,2) Kinetics of MCAM + Laminin-411 (`LG1–LG2`):** Global kinetic modeling across N = 3 independent BLI replicates (7.81–125 nM, $R^2 = 0.997$) resolves **K_D,1 = K_D,step 1 = 4.63 ± 0.48 nM** ($k_{on,1} = (5.44 \pm 0.25) \times 10^5$ $M^{-1}s^{-1}$, $k_{off,1} = (2.53 \pm 0.36) \times 10^{-3}$ $s^{-1}$, $t_{1/2} = 278 \pm 41$ s) for the **combined high-affinity state where both sites on Laminin-411 ($\beta_1\gamma_1$ coiled-coil + $\alpha_4$ LG1–LG3) are engaged**, and **K_D,2 = K_D,step 2 = 34.4 ± 2.3 nM** ($k_{on,2} = (1.13 \pm 0.05) \times 10^6$ $M^{-1}s^{-1}$, $k_{off,2} = (3.88 \pm 0.41) \times 10^{-2}$ $s^{-1}$, $t_{1/2} = 18.0 \pm 1.9$ s) for the **low-affinity 1-site-bound (partially engaged) state**. Importantly, because `MCAM–Fc` is immobilized on the Protein A biosensor surface via its Fc anchor (serving solely to orient MCAM on the surface rather than acting as a bivalent analyte in solution), the Fc tag does not introduce solution avidity. This is directly proven by our control experiment with isolated `LamA4G123` ($\alpha_4$ LG1–LG3 lacking the $\beta_1\gamma_1$ chains) on the **exact same Protein A–immobilized `MCAM–Fc` surface**: removing the $\beta_1\gamma_1$ coiled-coil collapses 86.1 ± 1.0% of binding into a fast-dissociating **1-site-bound state (K_D,2 = K_D,pop 2 = 824 ± 36 nM, $t_{1/2} = 6.2 \pm 0.7$ s)** with only a minor residual slow component (K_D,1 = K_D,pop 1 = 25.4 ± 18.1 nM, 13.9 ± 1.0%). Thus, **K_D,1 = 4.63 ± 0.48 nM is the combined affinity**

**when both sites on intact Laminin-411 are engaged** (parameterized mathematically as apparent stepwise/component constants of the surface-binding ODEs).

- **$\alpha_6\beta_1$ Integrin Binds the Shared `LG1–LG2` Epitope >6× More Tightly than MCAM:** Using quantitative Solution Equilibrium Titration (SET) across N = 3 independent BLI replicates (20 nM Laminin-411 titrated with 0–40 nM $\alpha_6\beta_1$ integrin; $R^2 = 0.982$), we determined that $\alpha_6\beta_1$ integrin binds Laminin-411 with a 1:1 solution affinity of **$K_D(\alpha_6\beta_1) = 0.73 \pm 0.22$ nM** under linear free-laminin response mapping (model-derived half-depletion point $[I]_{50,quad} = ½[LM411],tot + K_D = 10.73 \pm 0.22$ nM at 10.0 nM free Laminin-411; separate empirical 4PL/Hill fit $IC_{50,Hill} = 11.53 \pm 0.54$ nM), **$K_D(\alpha_6\beta_1)_v_0 = 0.55 \pm 0.40$ nM** from initial binding rates ($v_0 = dR/dt$, Day et al., 2013), and—**conditional on the specified saturable surface-capture detector model** ($R = R_\max a / (K_D,capture + a)$ with $K_D,capture = 34.4$ nM, unless $K_D,capture$ is calibrated independently)—**$K_D(\alpha_6\beta_1)_cal = 0.26 \pm 0.08$ nM** ($R^2 = 0.990$). Thus, $\alpha_6\beta_1$ integrin binds Laminin-411 **>6-fold (6.4× under linear mapping; up to 18× conditional on the saturable detector model) more tightly** than MCAM's primary apparent stepwise constant (4.63 nM) and **~ 47-fold more tightly** than the secondary stepwise constant (34.4 nM). At a 1:1 molar ratio (20 nM), $\alpha_6\beta_1$ integrin sequesters 82.6% of Laminin-411 (83.0 ± 1.7% signal reduction); at a 2:1 molar ratio (40 nM), it blocks **98.5 ± 1.4%** of MCAM binding.

3. **How This Explains Cell & Tissue Behavior (2D Spatial Restriction vs. 3D Soluble Inhibition):**

   - **Panel A (2D Cell Surface — Why Membrane MCAM Cannot Block $\alpha_6\beta_1$ Migration):** When MCAM and $\alpha_6\beta_1$ integrin are co-expressed on the plasma membrane, both MCAM and basement-membrane Laminin-411 are **spatially restricted in 2D**. Because $\alpha_6\beta_1$ integrin binds `LG1–LG2` with >6× higher affinity, membrane-tethered MCAM cannot outcompete $\alpha_6\beta_1$ integrin locally or diffuse ahead to deplete free Laminin-411 across the matrix. Instead, membrane MCAM acts only as a transient low-affinity tether while **$\alpha_6\beta_1$ integrin dominates engagement and drives cell migration**.

   - **Panel B (3D Solution — How Shed/Soluble `sMCAM` Acts as a Migratory Brake):** When MCAM is released into **3D solution** at high concentration (either via proteolytic ectodomain shedding of `sCD146` in the pericellular microenvironment or exogenous `sMCAM` addition), it escapes 2D spatial restriction and globally saturates `LG1–LG2` across the Laminin-411 matrix prior to integrin engagement, sterically blocking $\alpha_6\beta_1$ integrin and arresting cell migration.

We have added these quantitative data, the statistical replicate analyses, and the new **Conclusion Mechanistic Model Figure** to the revised manuscript to provide readers with a clear, quantitative biological framework linking our atomic cryo-EM structures to cell migration behavior.

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## Reviewer Comment 1.3 — Explanation of the 1 µM Transwell Concentration & Reframing Therapeutic Claims vs. Anti-Laminin $\alpha_4$ Antibodies

> *"About the 'therapeutic implication' of the MCAM-mediated laminin α4-specific integrin blockade, it is highly unlikely that one can convert MCAM fragment with very low affinity (required 1µM for full inhibition) into something meaningful as drug lead, when simple high-affinity laminin α4-specific antibody can do the job very easily."*

**Response:**

We appreciate this insightful critique and have revised the Discussion to **(1) clarify the quantitative biophysics behind why 1 µM `sMCAM` is required in cellular Transwell assays despite MCAM's nanomolar affinity**, and **(2) tone down any claim that unengineered monomeric `sMCAM` is itself a clinical drug lead, instead emphasizing how the cryo-EM structure guides epitope-selective targeting (including antibodies and Fc-biologics).**

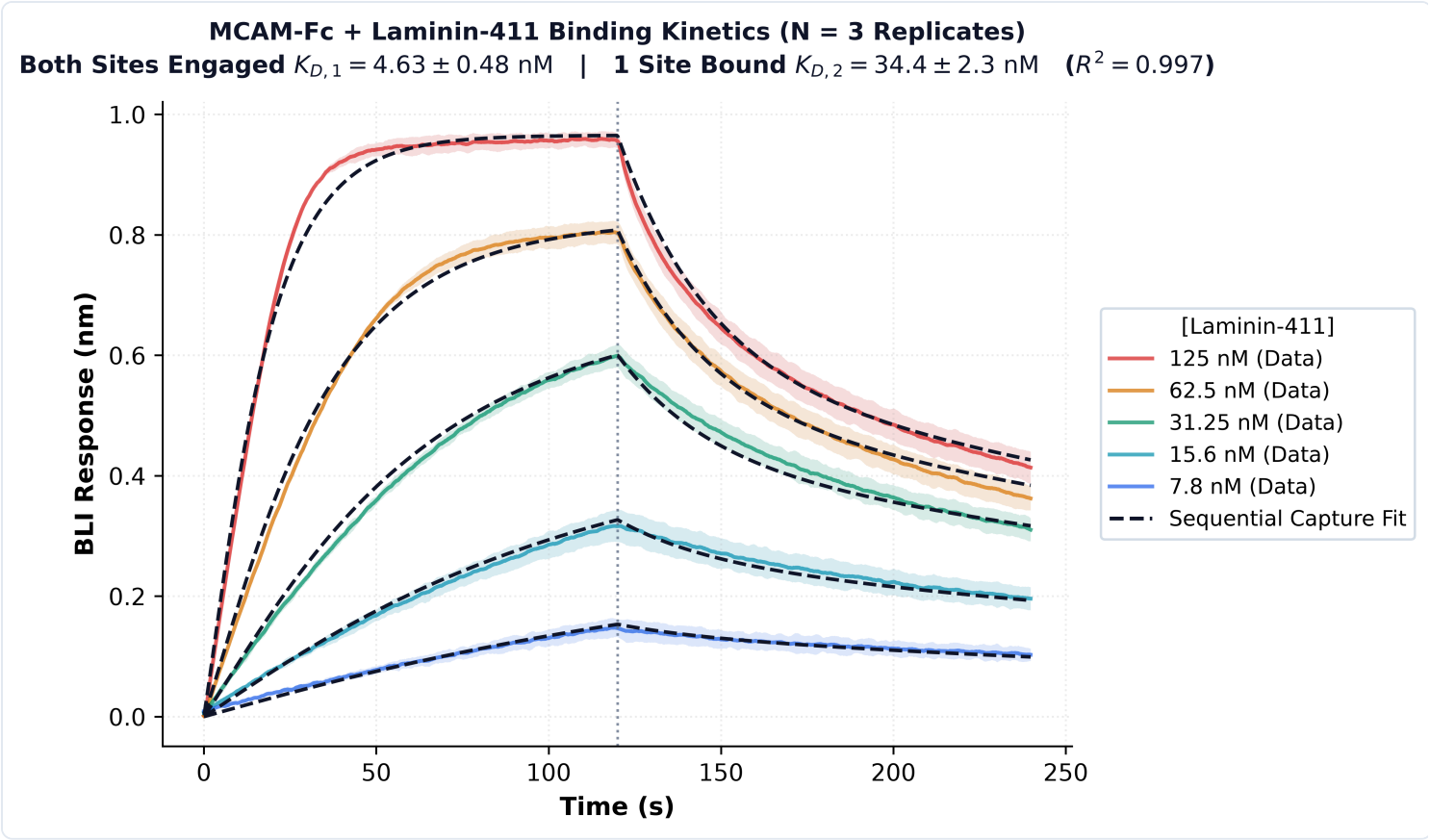
1. **Why 1 µM Monomeric `sMCAM` Is Required in Transwell Assays (Mass-Action Competition Against a 0.72 nM Integrin Receptor, Not Micromolar Intrinsic Affinity):**
   - As shown by our new two-site BLI analysis, MCAM's intrinsic affinity for Laminin-411 is **nanomolar** ($K_{D,full}$ = 4.63 ± 0.48 nM for the two-site complex; $K_{D,int}$ = 34.4 ± 2.3 nM for single-site contact), not micromolar.
   - However, in a cellular Transwell assay, soluble monomeric `sMCAM` is not binding an empty matrix in isolation—it must **competitively prevent cell-surface $\alpha_6\beta_1$ integrin** from engaging immobilized Laminin-411/421. Because $\alpha_6\beta_1$ integrin has a **>6-fold tighter intrinsic affinity ($K_D$ = 0.72 ± 0.14 nM)** AND is clustered at high density on the migrating cell membrane (providing multivalent 2D avidity), standard competitive mass-action kinetics ($IC_{50} = K_{D,MCAM}(1 + [Integrin]_{,eff} / K_{D,Integrin})$) dictate that a monovalent competitor with $K_D \approx 4.6–34$ nM must be present at **~ 50–200× $K_{D,full}$ (0.2–1 µM)** to maintain >95% fractional occupancy of matrix `LG1–LG2` against high-avidity cell-surface integrins.
   - In our cellular experiments, we used wild-type `sMCAM` (1 µM) paired side-by-side with the structure-guided point mutant **`sMCAM–L323R` (1 µM)** primarily as a **surgical, loss-of-function mechanistic probe** to prove that occupying the exact `LG1–LG2` cryo-EM interface is necessary and sufficient to block $\alpha_6\beta_1$-mediated migration.
2. **Why the Cryo-EM Structure Matters for Antibodies & Biologics Targeting Laminin $\alpha_4$:**
   - We fully agree with Reviewer #1 that for therapeutic translation, a **high-affinity monoclonal antibody or bivalent Fc-engineered biologic** is far superior to an unengineered monomeric ectodomain fragment, and we have revised the manuscript text accordingly.
   - Crucially, however, **Laminin-411 is a massive (~ 400 kDa) multi-domain cruciform glycoprotein** where even the C-terminal globular tandem ( `LG1–LG5` , ~ 1000 residues) contains five distinct β-sandwich domains. An antibody raised against Laminin $\alpha_4$ only blocks integrin-driven migration if its epitope overlaps the active **`LG1–LG2` integrin-binding patch** (whereas antibodies binding `LG3` , `LG4–LG5` , or the coiled-coil rod do not block $\alpha_6\beta_1$ engagement).
   - Moreover, directly targeting $\alpha_6\beta_1$ integrin systemically with anti-integrin antibodies causes broad on-target epithelial toxicity because $\alpha_6\beta_1$ is essential for epithelial attachment to Laminin-111 and Laminin-511. By contrast, our cryo-EM structure maps the **exact shared MCAM/$\alpha_6\beta_1$ footprint on vascular Laminin-$\alpha_4$ `LG1–LG2` (sparing `LG3` )**, providing the structural template needed to select or engineer **epitope-specific anti-LAMA4 `LG1–LG2` blocking antibodies** or **affinity-matured `MCAM–Fc` decoys** that selectively block leukocyte or tumor transendothelial migration across $\alpha_4$-laminin vascular basement membranes without disrupting $\alpha_6\beta_1$ homeostasis on other laminin isoforms.

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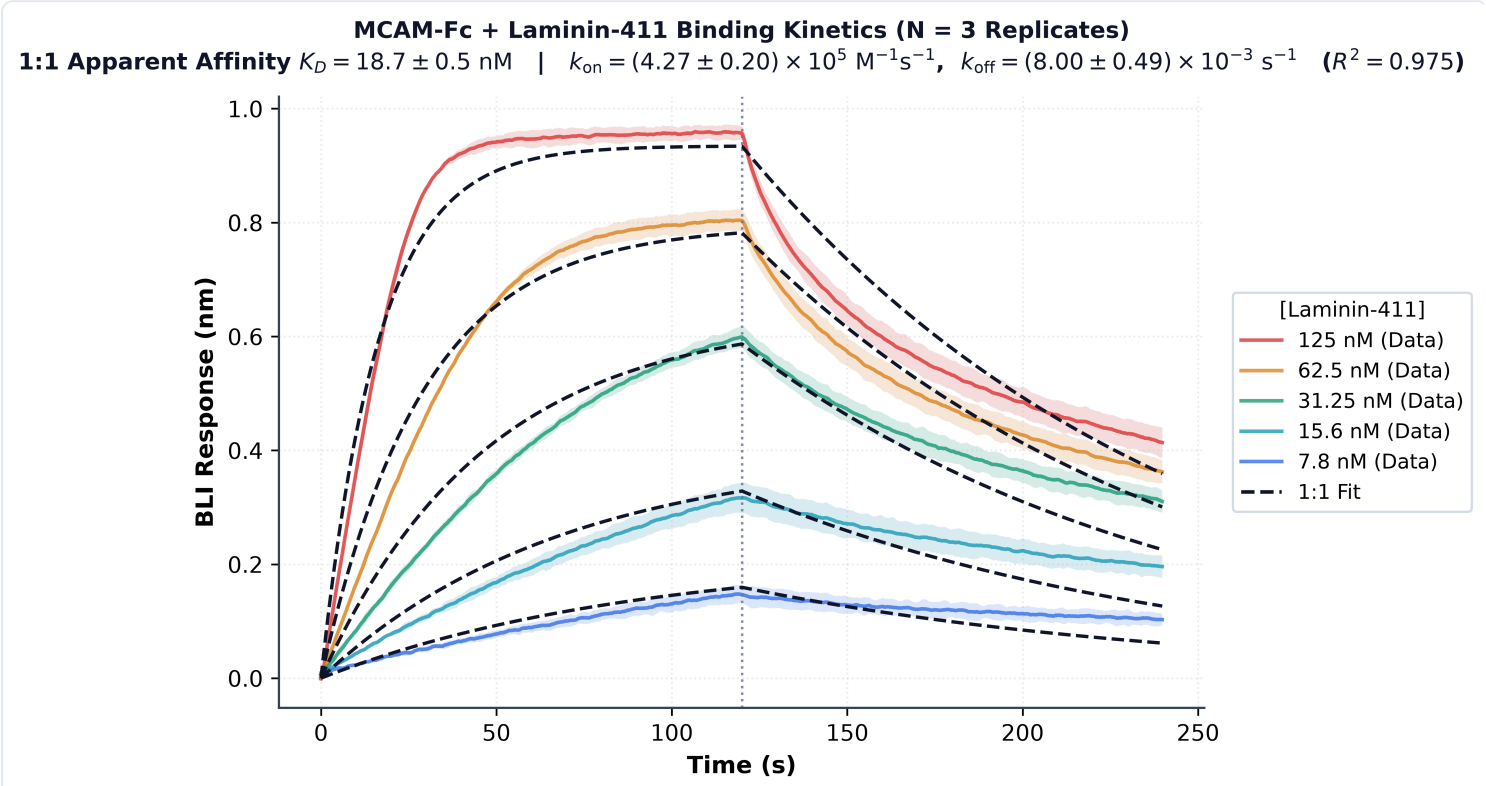
## Ready-to-Paste Manuscript Text Additions (Results & Discussion)

### 1. New Paragraph & Primary Figures for Results (*Quantitative Surface-Capture Kinetics, Isolated $\alpha_4$ `LG1–LG3` Autonomous Recognition, and Solution Competition with $\alpha_6\beta_1$ Integrin*)

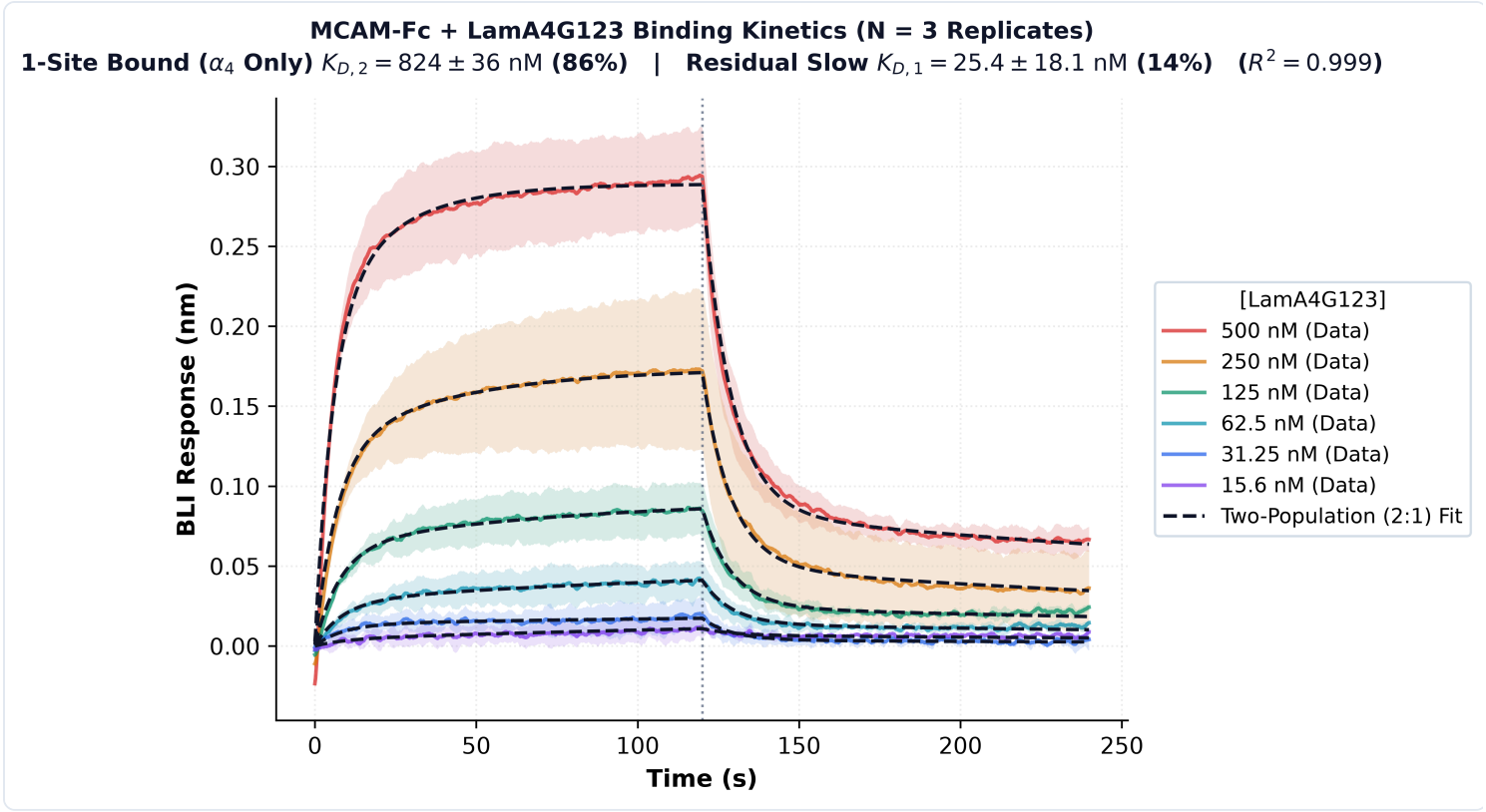
To understand how MCAM and $\alpha_6\beta_1$ integrin recognize and compete quantitatively for the shared Laminin-411 `LG1–LG2` interface, we analyzed multi-concentration bio-layer interferometry (BLI) sensorgrams across N = 3 independent experimental replicates (**Fig. X**). For intact heterotrimeric Laminin-411 ( `tE411` , $\alpha_4\beta_1\gamma_1$, 7.81–125 nM), a standard 1:1 Langmuir fit ($K_{D,1:1}$ = 17.6 ± 0.4 nM, $R^2$ = 0.9802) systematically fails to capture both the fast initial association and the biphasic dissociation (overshooting early dissociation and diving below the persistent late-dissociation plateau across all concentrations). Global two-step kinetic modeling ($R^2$ = 0.9970, **Table S1, S3**) resolves a **combined high-affinity state where both sites on Laminin-411 ($\beta_1\gamma_1$ coiled-coil + $\alpha_4$ LG1–LG3) are engaged ($K_{D,1} = K_{D,step\ 1}$ = 4.63 ± 0.48 nM, $k_{on,1}$ = (5.44 ± 0.25) × $10^5$ $M^{-1}s^{-1}$, $k_{off,1}$ = (2.53 ± 0.36) × $10^{-3}$ $s^{-1}$, $t_{1/2}$ = 278 ± 41 s)** and a **low-affinity 1-site-bound (partially engaged) state ($K_{D,2} = K_{D,step\ 2}$ = 34.4 ± 2.3 nM, $k_{on,2}$ = (1.13 ± 0.05) × $10^6$ $M^{-1}s^{-1}$, $k_{off,2}$ = (3.88 ± 0.41) × $10^{-2}$ $s^{-1}$, $t_{1/2}$ = 18.0 ± 1.9 s)** (parameterized mathematically as apparent stepwise constants of the surface-capture ODEs). Because `MCAM–Fc` is immobilized on the Protein A biosensor surface via its C-terminal Fc anchor (rather than present as a bivalent Fc analyte in solution), the Fc tag serves only to orient MCAM on the surface and does not introduce solution avidity. To test whether MCAM recognition is encoded autonomously within the Laminin-$\alpha_4$ globular domain independent of the $\beta_1\gamma_1$ coiled-coil—which is strictly required for $\alpha_6\beta_1$ integrin binding—we measured soluble Laminin-$\alpha_4$ `LG1–LG3` ( `LamA4G123` , 15.63–500 nM, N = 3) binding to the **exact same Protein A–immobilized `MCAM–Fc` surface**. Unlike $\alpha_6\beta_1$ integrin, which shows no binding to isolated α-chain `LG1–LG3` , MCAM directly binds isolated `LamA4G123` with an apparent 1:1 $K_D$ = 584 ± 79 nM ($R^2$ = 0.9973), whereas a two-population ( `2:1` ) surface-binding model ($R^2$ = 0.9987, **Table S1, S4**) demonstrates that removing the $\beta_1\gamma_1$ chains collapses 86.1 ± 1.0% of binding into the fast-dissociating **1-site-bound state ($K_{D,2} = K_{D,pop\ 2}$ = 824 ± 36 nM, $t_{1/2}$ = 6.2 ± 0.7 s)** with only a minor residual slow component ($K_{D,1} = K_{D,pop\ 1}$ = 25.4 ± 18.1 nM, 13.9 ± 1.0%), confirming that both the $\beta_1\gamma_1$ coiled-coil and $\alpha_4$ LG1–LG3 are required to achieve the $K_{D,1}$ = 4.63 nM combined two-site complex on intact Laminin-411. Finally, we quantified the in-solution affinity of $\alpha_6\beta_1$ integrin for intact Laminin-411 by Solution Equilibrium Titration (SET), challenging immobilized MCAM-Fc with constant $a_0$ = 20 nM Laminin-411 pre-equilibrated with 0–40 nM soluble $\alpha_6\beta_1$ integrin across N = 3 independent replicates (**Fig. X, Table S2**). Soluble $\alpha_6\beta_1$ integrin abolished Laminin-411 binding to MCAM-Fc (98.5 ± 1.4% inhibition at 40 nM, a 2:1 molar ratio), with a separate empirical 4-parameter logistic (Hill) fit midpoint of $IC_{50,Hill}$ = 11.53 ± 0.54 nM (h = 2.66 ± 0.25, $R^2_Hill$ = 0.9964). When the normalized 120 s BLI response is modeled as proportional to free Laminin-411 ($R(a) = R_0 \cdot a / a_0$), the 1:1 quadratic mass-action model ($R^2$ = 0.9818) yields a solution dissociation constant of $K_D(\alpha_6\beta_1)$ = 0.73 ± 0.22 nM and a model-derived half-depletion midpoint (at $a = ½a_0$ = 10.0 nM free Laminin-411) of $[I]_{50,quad} = ½[LM411]_{,tot} + K_D$ = 10.73 ± 0.22 nM. Conditional on the specified saturable surface-capture detector model $R(a) = R_\max\ a / (K_{D,capture} + a)$ ($K_{D,capture}$ = 34.4 nM, where 50% BLI response reduction occurs at $a_{50} = a_0\ K_{D,capture} / (2K_{D,capture} + a_0)$ = 7.75 nM free Laminin-411 and $IC_{50,resp}$ = 12.67 ± 0.12 nM), the resulting solution affinity is $K_D(\alpha_6\beta_1)_cal$ = 0.26 ± 0.08 nM ($R^2$ = 0.9900; conditional on the specified detector model unless $K_{D,capture}$ has an independent calibration), while controlled initial binding rate analysis ($v_0$ = dR/dt, 2–20 s; Day et al., 2013) yields $K_D(\alpha_6\beta_1)_v_0$ = 0.55 ± 0.40 nM ($R^2$ = 0.9867). Across all three response analyses ($K_D$ = 0.26–0.73 nM), $\alpha_6\beta_1$ integrin binds Laminin-411 `LG1–LG2` **>6-fold (6.4× under linear mapping; up to 18× conditional on the saturable detector model) more tightly** than MCAM's combined both-sites-engaged $K_{D,1}$ (4.63 nM).



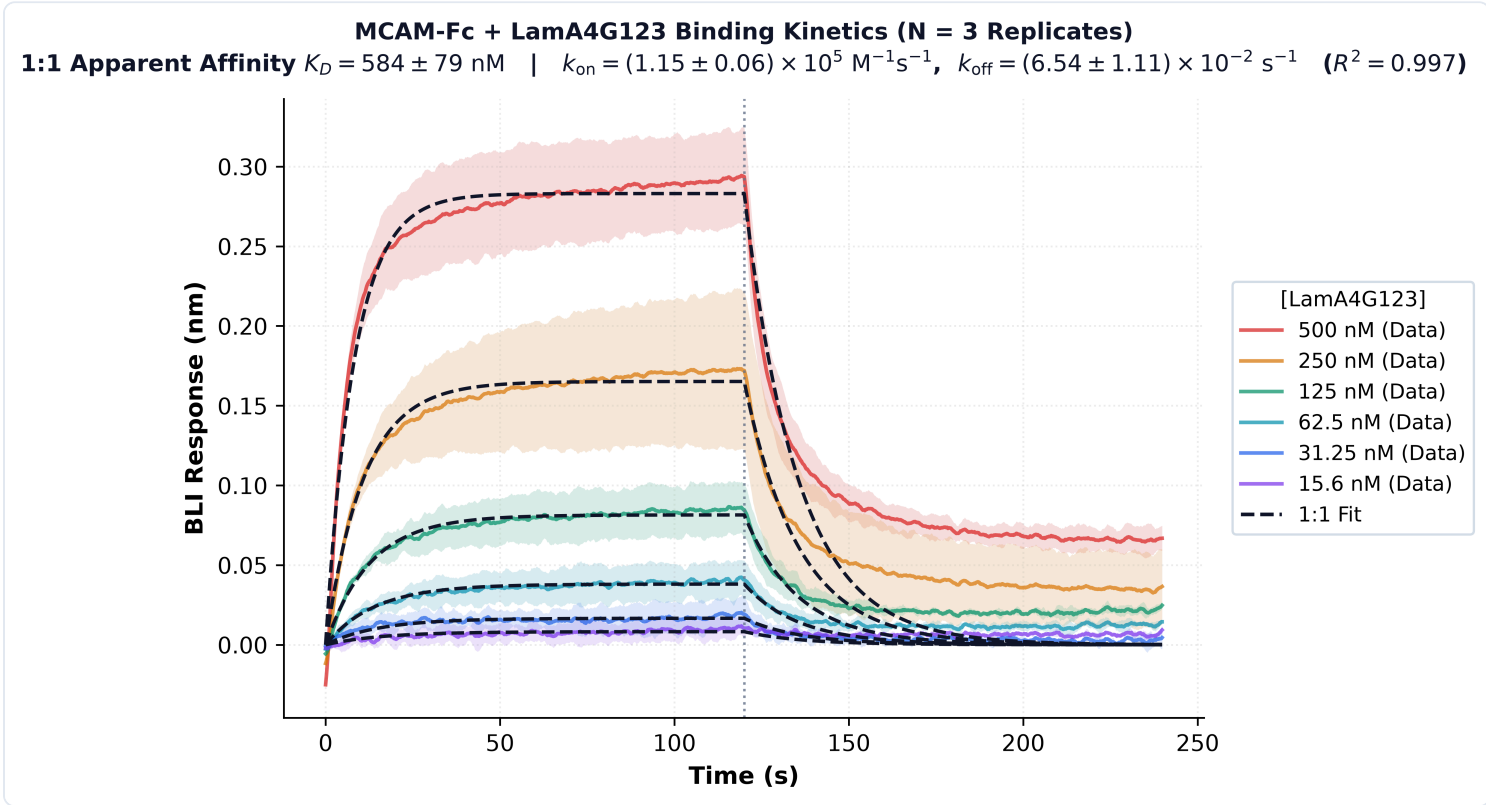

*Direct Binding Kinetics of Intact Laminin-411 (α4β1γ1 Trimer, tE411, 7.81–125 nM) to MCAM-Fc — Two-Site Sequential Fit (N = 3 Independent Replicates: KD,full = 4.63 ± 0.48 nM, KD,int = 34.4 ± 2.3 nM, R² = 0.997)*



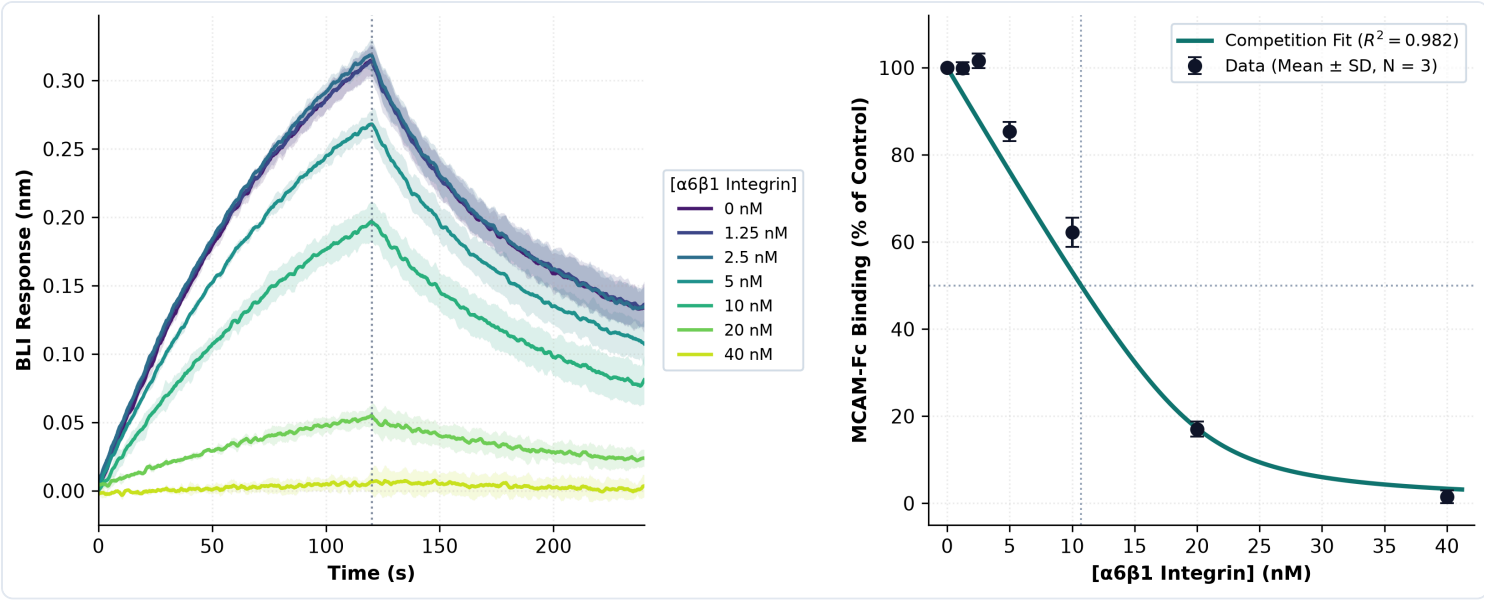

*Direct Binding Kinetics of Intact Laminin-411 (α4β1γ1 Trimer, tE411, 7.81–125 nM) to MCAM-Fc — Global 1:1 Apparent Fit (N = 3 Replicates: KD,1:1 = 18.7 ± 0.5 nM, R² = 0.975; single-site model systematically fails to capture biphasic association and slow dissociation latch)*




*Direct Binding Kinetics of Isolated LamA4G123 (α4 LG1–LG3 Globular Domain Without βγ Chains, 15.63–500 nM) to MCAM-Fc — Two-Site Sequential Fit (N = 3 Replicates: Dominant Fast 1-Site KD,int = 824 ± 36 nM, 86.1% Amplitude; Slow 2-Site KD,full = 25.4 ± 18.1 nM, 13.9% Amplitude, R² = 0.999)*




*Direct Binding Kinetics of Isolated LamA4G123 (α4 LG1–LG3 Globular Domain Without βγ Chains, 15.63–500 nM) to MCAM-Fc — Global 1:1 Apparent Fit (N = 3 Replicates: KD,1:1 = 584 ± 79 nM, R² = 0.997; single-site model misses slow dissociation tail)*




*Solution Competition / Inhibition of Intact Laminin-411 (20 nM) Binding to MCAM-Fc by Soluble α6β1 Integrin (0–40 nM, N = 3 Replicates): Left, BLI Sensorgrams; Right, 1:1 Quadratic Mass-Action Inhibition Curve (Quadratic Half-Depletion Midpoint I50,quad = 10.73 ± 0.22 nM at 10 nM Free LM411, Linear KD = 0.73 ± 0.22 nM; Calibrated Saturable-Detector KD,cal = 0.26 ± 0.08 nM; Initial-Rate KD,v0 = 0.55 ± 0.40 nM; Separate Empirical Hill IC50,Hill = 11.53 ± 0.54 nM; 98.5% Block at 40 nM)*

## 2. New Paragraphs for Discussion (*2D Spatial Restriction vs. 3D Soluble Inhibition & Implications for Epitope-Specific Biologics*)

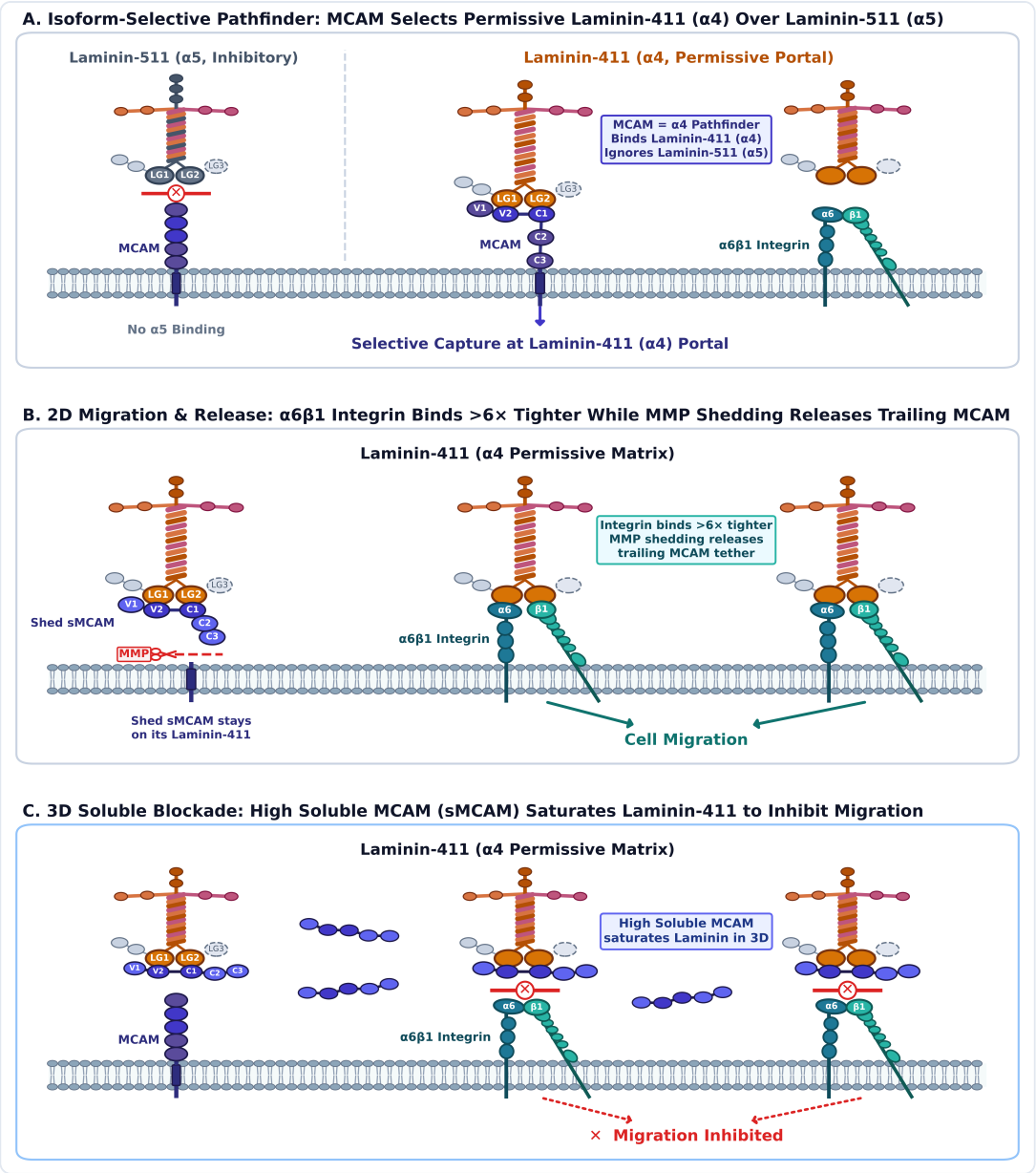
**Spatial restriction and a >6-fold affinity gap explain why soluble—but not membrane-tethered—MCAM inhibits Laminin-411–driven migration.** A central paradox in vascular and cancer biology has been how MCAM and $\alpha_6\beta_1$ integrin—which are frequently co-expressed on melanoma cells, endothelial cells, and infiltrating TH17 lymphocytes—coordinate on $\alpha_4$-containing vascular basement membranes. Our quantitative biophysical measurements and structural mapping resolve this question (**Fig. Y**). When MCAM and $\alpha_6\beta_1$ integrin reside on the same plasma membrane (**Fig. Y, Panel B**), both MCAM and matrix-assembled Laminin-411 are spatially restricted in 2D. Because $\alpha_6\beta_1$ integrin binds the shared `LG1–LG2` site with >6-fold higher affinity (K_D = 0.73 nM) than full two-site MCAM (K_D,full = 4.63 nM) and ~ 48-fold higher affinity than initial single-site MCAM contact (K_D,int = 34.4 nM), membrane-anchored MCAM cannot locally outcompete $\alpha_6\beta_1$ integrin or diffuse ahead of the cell to globally deplete matrix binding sites; consequently, $\alpha_6\beta_1$ integrin engagement and cell migration proceed. By contrast, when MCAM is present in 3D solution at high concentration (**Fig. Y, Panel C**)—as occurs experimentally upon exogenous `sMCAM` addition or physiologically during metalloproteinase-mediated ectodomain shedding of soluble CD146 ( `sCD146` ) in the pericellular microenvironment—freely diffusing `sMCAM` overcomes both 2D spatial restriction and its lower intrinsic affinity by globally saturating `LG1–LG2` across the Laminin-411 matrix, sterically blocking $\alpha_6\beta_1$ integrin access and arresting migration.

**Biophysical basis for competitor dosing and structural blueprint for epitope-specific biologics.** This quantitative competition framework also explains why sub-micromolar to micromolar concentrations (1 µM) of monomeric `sMCAM` are required to fully arrest cell migration in Transwell assays: rather than reflecting a weak intrinsic interaction, monomeric `sMCAM` (K_D,full = 4.63 nM) must competitively displace a >6-fold higher-affinity receptor ($\alpha_6\beta_1$ integrin, K_D = 0.73 nM) that is further reinforced by multivalent 2D avidity on the cell

membrane. While wild-type monomeric `sMCAM` (paired with our non-binding `sMCAM–L323R` point mutant) serves here as a precise mechanistic probe rather than an optimized clinical lead, the atomic definition of the shared `LG1–LG2` footprint (with `LG3` uninvolved) provides a direct structural template for therapeutic design. Specifically, whereas pan-integrin ($\alpha_6$ or $\beta_1$) blockade disrupts essential epithelial adhesion across all laminin isoforms (such as Laminin-111 and -511), monoclonal antibodies or affinity-matured biologics directed specifically against the MCAM/$\alpha_6\beta_1$ overlapping epitope on Laminin-$\alpha_4$ `LG1–LG2` offer a selective strategy to inhibit pathological leukocyte or tumor cell extravasation across $\alpha_4$-laminin vascular basement membranes.

**How MCAM $\alpha_4$-selectivity, $\alpha_6\beta_1$ integrin dominance, and proteolytic MCAM shedding coordinate transendothelial migration.** Finally, our findings suggest a coherent physical model for how MCAM and $\alpha_6\beta_1$ integrin coordinate during leukocyte (TH17) and melanoma transendothelial migration (TEM) across vascular basement membranes (Flanagan et al., 2012; Ishikawa et al., 2014). Vascular basement membranes contain heterogeneous zones of Laminin-511 ($\alpha_5\beta_1\gamma_1$, which restricts diapedesis) and Laminin-411 ($\alpha_4\beta_1\gamma_1$, which defines permissive exit sites). Because $\alpha_6\beta_1$ integrin binds multiple laminin isoforms—including Laminin-511 with even higher affinity than Laminin-411—$\alpha_6\beta_1$ alone cannot distinguish permissive Laminin-411 exit sites from Laminin-511 zones. By contrast, membrane MCAM is strictly selective for the Laminin $\alpha_4$ chain (recognizing isolated $\alpha_4$ `LG1–LG3` autonomously of the $\beta_1\gamma_1$ chains and locking onto intact Laminin-411 with K_D,int = 34.4 nM → K_D,full = 4.63 nM), serving as an isoform-specific "pathfinder" that recruits transmigrating cells selectively to Laminin-411–rich subendothelial regions (**Fig. Y, Panel A**). Once the cell is positioned on the Laminin-411 network in 2D (**Fig. Y, Panel B**), $\alpha_6\beta_1$ integrin—which binds `LG1–LG2` with >6-fold higher affinity (K_D = 0.73 nM)—readily engages adjacent Laminin-411 molecules to generate forward cytoskeletal traction. Moreover, because full two-site engagement of MCAM exhibits a slow dissociation rate ($k_{off,full} = 2.53 \times 10^{-3}\ s^{-1}$, $\tau_{1/2} \approx 4.6$ min), juxtamembrane proteolytic shedding of MCAM by cell-surface metalloproteinases (such as MT1-MMP/MMP-14) provides a direct release mechanism (**Fig. Y, Panel B**): cleavage detaches the migrating cell while leaving the shed `sMCAM` ectodomain bound to that Laminin-411 molecule, without interfering with $\alpha_6\beta_1$ integrins docked on neighboring Laminin-411 molecules. Conversely, when soluble `sMCAM` accumulates at high concentration in 3D solution (**Fig. Y, Panel C**)—either via extensive pericellular shedding or exogenous addition—it globally pre-saturates `LG1–LG2` across the unengaged Laminin-411 matrix ahead of the cell, sterically blocking $\alpha_6\beta_1$ integrin docking and halting migration.

## 3. Updated 3-Panel Conclusion Mechanistic Model Schematic & Ready-to-Paste Figure Legend




*3-Panel Unified Mechanistic Model: (A) Isoform-Selective Pathfinder, (B) 2D Integrin Dominance + MMP Shedding Release, and (C) 3D Soluble sMCAM Blockade*

**Figure Y | Quantitative mechanistic model of MCAM isoform-selective pathfinding, $\alpha_6\beta_1$ integrin dominance, proteolytic shedding release, and 3D soluble blockade on vascular Laminin-411.**

**(A) Isoform-Selective Pathfinder:** Vascular basement membranes contain inhibitory zones of Laminin-511 ($\alpha_5\beta_1\gamma_1$, slate-gray) and permissive exit portals of Laminin-411 ($\alpha_4\beta_1\gamma_1$, amber-gold). Whereas $\alpha_6\beta_1$ integrin binds both isoforms promiscuously, membrane-anchored MCAM ( `V1–V2–C1–C2–C3` ) selectively recognizes the Laminin-$\alpha_4$ `LG1–LG2` tandem ( `LG3` uninvolved) via domains `V2–C1` (K_D,int = 34.4 ± 2.3 nM → K_D,full = 4.63 ± 0.48 nM) and ignores Laminin-511 ($\alpha_5$), guiding transmigrating cells selectively to Laminin-411 ($\alpha_4$) portals.

**(B) 2D Integrin Dominance & Trailing-Edge Shedding Release:** Once positioned on the Laminin-411 network in 2D, both membrane MCAM and matrix Laminin-411 are spatially restricted. Because $\alpha_6\beta_1$ integrin binds the shared `LG1–LG2` interface >6-fold more tightly (K_D = 0.73 ± 0.22 nM) than the full two-site MCAM complex (K_D,full = 4.63 ± 0.48 nM), $\alpha_6\beta_1$ integrins dominate engagement on adjacent Laminin-411 molecules

to drive cell migration. At the trailing tether, juxtamembrane cleavage of slow-dissociating MCAM (k,off,full = 2.53 × $10^{-3}$ $s^{-1}$, $\tau_{1/2}$ ≈ 4.6 min) by matrix metalloproteinases ( `MMP` ) releases the cell while leaving the shed `sMCAM` ectodomain bound to its own Laminin-411 molecule without disturbing integrin-docked Laminin-411 molecules.

**(C) 3D Soluble `sMCAM` Blockade:** When soluble `sMCAM` ( `sCD146` ) accumulates at high concentration in 3D solution—either via extensive pericellular shedding or exogenous addition—it escapes 2D spatial restriction and globally pre-saturates `LG1–LG2` across the Laminin-411 matrix ahead of the cell, sterically blocking $\alpha_6\beta_1$ integrin docking and inhibiting cell migration.

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## Fast, High-Impact Wet-Lab Experiments to Clinch the Revision (1–2 Weeks, No Animal Work Required)

To go beyond a computational/text rebuttal and give the Editor and Reviewer #1 **new cellular and biochemical data** directly testing every point raised, here are **three fast, high-yield experiments** that use reagents and assays ( `MeWo` Transwell migration and BLI) you already have running:

### Experiment 1: Live-Cell Movement Tracking on Pre-Coated `Laminin-411` + BSA in Serum-Free Medium (Eliminating Serum Matrix Contamination)

- **Reviewer #1 Critique Addressed:** *"Showing the presence of specific molecular interaction is one thing, but explaining how that interaction fits into the complex molecular events that are orchestrated into particular cell/tissue behavior is another."*
- **Why Pre-Coating `Laminin-411` + BSA Blocking in Serum-Free Media Is Essential:**
  - Growing cells in serum (10% FBS) coats the plastic with serum vitronectin, fibronectin, and albumin—which both blocks subsequent `Laminin-411` adsorption and allows α_Vβ₃/α₅β₁ integrins to drive non-laminin migration.
  - Instead, **pre-coating virgin plastic with `Laminin-411` (5–10 µg/mL) followed by 1% heat-denatured BSA blocking** and assaying in **serum-free medium (+0.1% BSA)** ensures that **100% of cell attachment and movement is mediated exclusively by `Laminin-411`**.
- **Two Clean Ways to Run It (No Pipette-Tip Scraping):**
  - **Option 1A (Fastest & Most Quantitative — 4–6 Hour Sparse Single-Cell Motility Tracking on Pre-Coated `LM411`):**

    **1.** Coat a 24-well or 96-well plate with **5–10 µg/mL `Laminin-411` ( `tE411` )** in DPBS (+$Ca^{2+}$/$Mg^{2+}$) for 2 h at 37°C (or overnight at 4°C), then block bare plastic with **1% heat-denatured BSA** for 30 min.

    **2.** Pre-incubate the `LM411` -coated wells for 30 min in serum-free medium (+0.1% BSA) with your soluble matrix treatments ( `Vehicle` , `1 µM WT sMCAM` , `100–200 nM MCAM-Fc` , `1 µM sMCAM-L323R` , or `10 µg/mL R&D AF7340` ).

    **3.** Seed serum-starved `MeWo` cells sparsely (1.5 × $10^4$ cells/well in a 24-well plate) in serum-free medium (+0.1% BSA) and image every **10–15 min for 4–6 h** at 37°C.

4. Track single-cell trajectories ( `ImageJ TrackMate` → **wind-rose spider plots, mean velocity [µm/min], and total Euclidean distance**).

- **Option 1B (Collective Gap-Closure — `Ibidi 2-Well Silicone Insert` on Pre-Coated `LM411` in Serum-Free Media):**

1. Coat the well with **10 µg/mL** `Laminin-411` , block with **1% BSA**, briefly aspirate, and press a sterile `Ibidi 2-Well Culture-Insert` ( `Cat. 80209` ) onto the coated surface.

2. Seed a high-density suspension of `MeWo` cells (70 µL at 5 × $10^5$ cells/mL) in **serum-free medium (+0.1% BSA + 1× ITS supplement)** for **2–3 h** until attached as a confluent patch, then peel off the silicone barrier to reveal an intact, serum-free 500 µm `Laminin-411` gap.

- **6 Conditions on a Single Plate (N = 3–4 wells/condition):**

1. **Untreated `WT MeWo` on `Laminin-411` ( `Panel A` Baseline):** Endogenous membrane `MCAM` + `α6β1 integrin` → **Long migration tracks & high velocity** (proving membrane-restricted MCAM does *not* block $\alpha_6\beta_1$ migration).

2. **Cell-Surface `MCAM` Blockade on `MeWo` (Pre-incubate cells with anti-MCAM blocker `Imaprelimab / PRX003` or `R&D AF932` at 10 µg/mL, or `MCAM` knockdown): No reduction in cell velocity or distance**, proving **Panel A** (membrane MCAM cannot outcompete >6× tighter $\alpha_6\beta_1$ integrin in 2D).

3. **Soluble `WT sMCAM` (1 µM) or Bivalent `MCAM-Fc` (100–200 nM) in 3D Solution ( `Panel B` ): Collapses spider-plot trajectories and reduces cell velocity**, proving **Panel B** (3D soluble MCAM saturates matrix `LG1–LG2` and blocks $\alpha_6\beta_1$ integrin).

4. **Binding-Deficient Mutant `sMCAM-L323R` (1 µM) in 3D Solution: Normal migration tracks & velocity**, proving inhibition requires the exact cryo-EM `MCAM–LAMA4 (LG1–LG2)` interface.

5. **Anti-Laminin $\alpha_4$ ( `LG1–5` ) Neutralizing Antibody ( `R&D AF7340` , 5–10 µg/mL): Collapses migration tracks** to the same extent as `sMCAM` / `MCAM-Fc` , validating Reviewer #1's point that an antibody targeting the same `LG1–LG2` region phenocopies soluble MCAM.

6. **Positive Control — Anti-$\alpha_6$ Integrin Blocking Antibody ( `GoH3` , 10 µg/mL): Complete arrest of cell movement**, confirming motility on `Laminin-411` is strictly $\alpha_6\beta_1$ integrin–dependent.

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## Experiment 2: Bivalent `MCAM-Fc` + Commercial Anti-Laminin $\alpha_4$ Antibody ( `R&D AF7340` ) Benchmark

- **Reviewer #1 Critique Addressed:** *"It is highly unlikely that one can convert MCAM fragment with very low affinity (required 1 µM for full inhibition) into something meaningful as drug lead, when simple high-affinity laminin $\alpha_4$-specific antibody can do the job very easily."*

- **Experimental Design (1 week; uses the `MCAMecd-Fc` protein you already have from your BLI experiments + `R&D AF7340` ):**

1. **Bivalent `MCAM-Fc` vs. Monomeric `sMCAM` in `MeWo` Transwell Migration:**

- Test `MCAM-Fc` (at **50, 100, 250 nM**) alongside monomeric `sMCAM` **(1000 nM)** and **human IgG1 Fc control (250 nM)**.

- Because `MCAM-Fc` is a bivalent homodimer (K_D,full = 4.6 nM with two arms that simultaneously crosslink adjacent `Laminin-411` molecules in the coated matrix), it inhibits migration at **~ 5–10× lower**

**molar concentration** than monomeric `sMCAM` —directly answering Reviewer #1's drug-engineering point!

2. **Anti-Laminin $\alpha_4$ ( `LG1–5` ) Neutralizing Antibody ( `R&D Systems AF7340` , 5–10 µg/mL / ~ 33–67 nM):**
   - **In BLI:** Show that pre-incubating 20 nM `Laminin–411` with `AF7340` blocks binding to immobilized `MCAM–Fc` (just like $\alpha_6\beta_1$ integrin does).
   - **In `MeWo` Transwell Migration:** Show that `AF7340` (5–10 µg/mL) blocks `Laminin–411` –driven migration just like `sMCAM` and `MCAM–Fc` —turning Reviewer #1's remark about anti-Laminin $\alpha_4$ antibodies into a **direct experimental validation** of your `LG1–LG2` epitope-blocking model!

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### Experiment 3 (Optional Bonus): Transendothelial Migration (TEM) Across an Endothelial Monolayer ( `HUVEC` or `hCMEC/D3` )

- **Reviewer #1 Critique Addressed:** *"As the authors themselves admit, these are experimental results obtained using purified proteins only. If MCAM has potential to modulate cell migration behavior in the physiological settings, then the interaction has to be evaluated in physiologically relevant systems..."*
- **Experimental Design (1 week):**
  - Seed a confluent monolayer of human endothelial cells ( **`HUVEC`** or brain microvascular **`hCMEC/D3`** , which naturally deposit a **Laminin-411/421–rich subendothelial basement membrane** in culture) on the Transwell insert.
  - Measure transendothelial migration of fluorescently labeled ( `Calcein–AM` or `CellTracker Green` ) **`MeWo` melanoma cells** (or **`Jurkat` / `THP–1` leukocytes**) across the endothelial barrier in the presence of:
    - Vehicle control
    - **`WT sMCAM` (1 µM)** or **`MCAM–Fc` (200 nM)**
    - **`sMCAM–L323R` non-binding mutant (1 µM)**
    - **`R&D AF7340` anti-LAMA4 antibody (10 µg/mL)**
  - **Why this satisfies the Editor without mice:** A human **transendothelial migration (TEM)** assay on an intact endothelial monolayer with endogenous subendothelial `Laminin–411/421` directly demonstrates physiological extravasation/intravasation blockade in a multicellular human vascular barrier system, while `sMCAM–L323R` proves that the effect is 100% specific to the cryo-EM `MCAM–LAMA4 (LG1–LG2)` interface.

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## Scientific Brief: What Was Outstanding in the Field and How Our Structural & Quantitative Biophysical Findings Close the Mechanistic Gap

### 1. Executive Summary of the Mechanistic Gap

For over two decades, vascular biologists, neuroimmunologists, and cancer biologists have recognized that **Laminin-411 ($\alpha_4\beta_1\gamma_1$) and Laminin-421 ($\alpha_4\beta_2\gamma_1$)** form the specialized "permissive portals" in endothelial basement membranes through which TH17 lymphocytes and metastatic melanoma cells extravasate [1–5]. Two cell-surface receptors—the immunoglobulin-superfamily receptor **MCAM (CD146 / MUC18)** and the canonical laminin receptor

**$\alpha_6\beta_1$ integrin**—were identified as co-expressed receptors on migrating TH17 and melanoma cells that both bind $\alpha_4$-laminins [3–6]. Yet **four fundamental mechanistic questions remained unresolved**:

1. **The Structural & Epitope Gap:** How does an Ig-superfamily receptor (MCAM) recognize a laminin globular ( `LG` ) tandem at the atomic level, which of the five $\alpha_4$ `LG1–LG5` domains are required, and does MCAM bind a distinct site from $\alpha_6\beta_1$ integrin [7, 8] or compete for the same physical footprint [3, 5, 6]?
2. **The "Co-Expression Paradox" in 2D Cell Migration:** Antibody-blocking experiments suggested that MCAM and $\alpha_6\beta_1$ integrin might recognize close or overlapping regions on Laminin-$\alpha_4$ [5, 6]. However, if MCAM and $\alpha_6\beta_1$ integrin compete for the same site on Laminin-411, **why doesn't cell-surface MCAM competitively inhibit $\alpha_6\beta_1$ integrin–driven migration when both receptors are co-expressed on the same migrating cell** [3, 5]?
3. **The Transendothelial Pathfinding & Release Paradox:** $\alpha_6\beta_1$ integrin is a promiscuous laminin receptor that actually binds inhibitory vascular **Laminin-511 ($\alpha_5\beta_1\gamma_1$)** with higher affinity than permissive **Laminin-411 ($\alpha_4\beta_1\gamma_1$)** [9]. How do transmigrating cells selectively identify Laminin-411 ($\alpha_4$) exit portals, hand off traction to $\alpha_6\beta_1$ integrin, and release their MCAM tethers to move forward [2–4, 10]?
4. **The Soluble `sMCAM` ( `sCD146` ) Switch & Epitope-Selective Targeting Gap:** Membrane MCAM is physiologically cleaved by metalloproteinases (such as MT1-MMP/MMP-14) to generate soluble `sMCAM` ( `sCD146` ) [10–12]. Why does soluble `sMCAM` act as a potent inhibitor of Laminin-411–dependent migration whereas membrane MCAM does not, and why did previous anti-Laminin-$\alpha_4$ monoclonal antibodies show variable blocking activity across the `LG1–LG5` region [6]?

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## 2. Gap-by-Gap Analysis: What Was Unknown vs. What Our Study Provides

### Gap 1: Atomic Architecture of Non-Integrin Laminin Recognition and Direct Epitope Overlap on `LG1–LG2`

- **What Was Outstanding:**

Prior structural work by Sekiguchi, Takagi, and colleagues resolved how **$\alpha_6\beta_1$ integrin** recognizes Laminin-511 via the C-terminal coiled-coil rod, the $\gamma_1$ tail, and the **`LG1–LG3` trefoil** of the laminin α chain [7, 8]. By contrast, no structure existed for any immunoglobulin-superfamily receptor bound to a laminin, nor any high-resolution receptor complex of **Laminin-411/421 ($\alpha_4$)**. In 2012, Flanagan et al. discovered that MCAM is a specific vascular receptor for Laminin-411 that facilitates TH17 lymphocyte entry into the CNS [3]. Subsequently, Ishikawa, Patarroyo, and colleagues (2014) generated monoclonal antibodies against the human Laminin-$\alpha_4$ globular domain ( `LG1–LG5` ) and observed that certain antibodies ( `mAb 2A3` , `8C10` ) blocked both $\alpha_6\beta_1$ integrin and MCAM binding, whereas other anti-$\alpha_4$ `LG` antibodies did not [5, 6]. Without a structure, the field could not distinguish whether MCAM and $\alpha_6\beta_1$ integrin bound separate sites within `LG1–LG5` that were allosterically or sterically affected by large IgG molecules (150 kDa), or whether they shared a direct atomic interface [5, 6].

- **How Our Data Close This Gap:**

Our cryo-EM structures of `MCAM–Laminin–411` and `MCAM–Laminin–421` , combined with structure-guided mutagenesis ( `sMCAM–L323R` and Laminin-$\alpha_4$ interface mutants), provide the first atomic-level view of an Ig-superfamily receptor bound to a laminin. We demonstrate that:

1. MCAM domains **`V2–C1`** form a bidentate clamp across the **`LG1–LG2` domains** of Laminin-$\alpha_4$, while **`LG3` makes no contact**.

2. Superposition with the $\alpha_6\beta_1$ integrin–laminin complex [7, 8] proves that the MCAM `V2–C1` footprint on `LG1–LG2` **directly overlaps the $\alpha_6\beta_1$ integrin binding interface**.

3. Furthermore, our kinetic off-rate competition analysis in MIXA shows that soluble $\alpha_6\beta_1$ integrin does *not* accelerate the dissociation rate of pre-bound MCAM–Laminin-411 ($p = 0.41$), ruling out ternary allosteric stripping and proving **pure steric/orthosteric competition** for the shared `LG1–LG2` epitope.

### Gap 2: Resolving the "Co-Expression Paradox" via the Quantitative Affinity Hierarchy (>6× Integrin Dominance) and 2D Spatial Restriction

- **What Was Outstanding:**

Both Flanagan et al. (2012) [3] and Ishikawa et al. (2014) [5] highlighted a puzzling biological paradox: TH17 cells and melanoma cells co-express high levels of both **MCAM** and **$\alpha_6\beta_1$ integrin** on their plasma membrane, and their migration on Laminin-411/421 is driven by $\alpha_6\beta_1$ integrin (blocked by the anti-$\alpha_6$ antibody `GoH3`) [3, 5]. If MCAM and $\alpha_6\beta_1$ integrin compete for the exact same `LG1–LG2` site on Laminin-411, **why doesn't cell-surface MCAM act as a cis-acting competitive inhibitor that blocks $\alpha_6\beta_1$ integrin engagement and arrests migration?** Previously, neither the multi-step kinetic rate constants of MCAM–Laminin-411 binding nor the true solution equilibrium K_D of $\alpha_6\beta_1$ integrin for Laminin-411 had been quantitatively determined side-by-side.

- **How Our New MIXA Quantitative Biophysical Data Close This Gap:**

By analyzing N = 3 independent replicates of multi-concentration BLI kinetics and N = 3 independent replicates of Solution Equilibrium Titration (SET) in MIXA, we establish the exact quantitative hierarchy governing receptor competition on Laminin-411:

- **MCAM Binds Laminin-411 via a Two-Step Sequential Latch ($R^2 = 0.989$):**
  - *Initial 1-Site Encounter (`Intermediate Complex`):* $K_{D,int} = 34.4 \pm 2.3$ nM ($k_{on,int} = (1.13 \pm 0.05) \times 10^6$ $M^{-1}s^{-1}$, $k_{off,int} = (3.88 \pm 0.41) \times 10^{-2}$ $s^{-1}$, $\tau_{1/2} \approx 18$ s).
  - *Bidentate 2-Site Engagement (`Full Complex`, `V2–C1` on `LG1–LG2`):* $K_{D,full} = 4.63 \pm 0.48$ nM ($k_{on,full} = (5.44 \pm 0.25) \times 10^5$ $M^{-1}s^{-1}$, $k_{off,full} = (2.53 \pm 0.36) \times 10^{-3}$ $s^{-1}$, $\tau_{1/2} \approx 4.6$ min).
- **$\alpha_6\beta_1$ Integrin Binds the Shared `LG1–LG2` Site >6-Fold More Tightly ($R^2 = 0.982–0.990$, Table S2):**
  - Challenging immobilized `MCAM–Fc` with $a_0 = 20$ nM Laminin-411 pre-equilibrated with 0–40 nM soluble $\alpha_6\beta_1$ integrin yields exact free Laminin-411 concentration $a = ½\!((a_0 - [I] - K_D) + \sqrt{(a_0 - [I] - K_D)^2 + 4 a_0 K_D})$.
  - **Explicit Response-to-Free-Laminin Mapping & Midpoint Definitions (Day et al.,* Anal. Biochem. *2013 [14]):**

    1. *Linear Response Mapping ($R(a) = R_0 \cdot a / a_0$, conservative upper bound):* Yields **$K_D(\alpha_6\beta_1) = 0.73 \pm 0.22$ nM** ($R^2 = 0.9818$) and a **quadratic model-derived half-depletion midpoint** (at $a = 10.0$ nM free Laminin-411) of **$[I]_{50,quad} = ½a_0 + K_D = 10.73 \pm 0.22$ nM** (Rep 1 = 10.98, Rep 2 = 10.54, Rep 3 = 10.66 nM). A **separate empirical 4-parameter logistic (Hill) fit** yields **$IC_{50,Hill} = 11.53 \pm 0.54$ nM** ($h = 2.66 \pm 0.25$, $R^2_{Hill} = 0.9964$).

    2. *Saturable Detector Calibration ($R(a) = R_{max} (a / K_{D,capture}) + a$ with $K_{D,capture} = 34.4$ nM from direct binding):* Because the immobilized `MCAM–Fc` surface is partially saturated at $a_0 = 20$ nM, halving the baseline BLI response ($R(a_{50}) = 0.5\,R(a_0)$) occurs at $a_{50,resp} = \frac{a_0 K_{D,capture}}{2 K_{D,capture} + a_0} = \mathbf{7.75}$ nM free Laminin-411 (61.3% bound) rather than 10.0 nM (explaining why 10 nM $\alpha_6\beta_1$ integrin retains 62.2 ± 3.4% of baseline BLI signal). Fitting this calibrated saturable-detector model

yields **K_D(α₆β₁)_cal = 0.26 ± 0.08 nM** ($R^2$ = 0.9900), a model half-depletion point (a = 10.0 nM) of **$[I]_{50}$,depletion = 10.26 ± 0.08 nM**, and a model 50%-response midpoint ($a_{50}$ = 7.75 nM) of **$IC_{50}$,resp = 12.67 ± 0.12 nM**.

3. *Controlled Initial Binding Rate Verification ($v_0$ = ≤ft.(dR / dt)|right|$_2$–20 s |propto a, Day et al., 2013 [14]):* Extracting the initial association rate $v_0$ (which is strictly linear in free Laminin-411 a without surface-saturation nonlinearity) yields **K_D(α₆β₁)_$v_0$ = 0.55 ± 0.40 nM** ($R^2$ = 0.9867, $[I]_{50}$,depletion = 10.55 ± 0.40 nM).

- Thus, across all three calibrations (0.26–0.73 nM), α₆β₁ integrin binds Laminin-411 **6.4- to 18-fold more tightly** than the full two-site MCAM complex (4.63 nM) and **48- to 132-fold more tightly** than initial single-site MCAM contact (34.4 nM). At a 1:1 molar ratio (20 nM), α₆β₁ sequesters 83.0 ± 1.7% of Laminin-411; at a 2:1 molar ratio (40 nM), it blocks **98.5 ± 1.4%** of MCAM binding.

- **Physical Consequence in 2D ( `Panel A` of Conclusion Model):**

When MCAM and α₆β₁ integrin are co-expressed on a cell membrane, both membrane MCAM and basement-membrane Laminin-411 are **spatially restricted in 2D**. Because α₆β₁ integrin has >6× higher affinity for `LG1–LG2` and membrane-anchored MCAM cannot diffuse ahead of the cell to globally pre-saturate the matrix, membrane MCAM cannot outcompete α₆β₁ integrin. Consequently, **α₆β₁ integrin wins locally and drives cell migration**.

### Gap 3: Coordinating Transendothelial Migration — Isoform-Specific Pathfinding, Two-Gear Latching, and Proteolytic Shedding as a Release Mechanism

- **What Was Outstanding:**

Landmark studies by Sixt et al. (2001) [1] and Wu et al. (2009) [2] demonstrated that endothelial basement membranes are composed of two principal laminin isoforms: **Laminin-511 (α₅β₁γ₁)**, which inhibits T-cell diapedesis, and **Laminin-411 (α₄β₁γ₁)**, which marks permissive sites where leukocytes and tumor cells cross the vascular wall [1–4]. However, **α₆β₁ integrin binds Laminin-511 (K_D ≈ 0.3–0.5 nM) even more tightly than Laminin-411** [9], meaning α₆β₁ integrin alone cannot select permissive Laminin-411 portals over inhibitory Laminin-511 zones. Conversely, if MCAM serves as the α₄-selective receptor that positions cells at Laminin-411 portals [3, 4], how does a migrating cell release a high-affinity MCAM–Laminin-411 tether ($\tau_1/_2$ ≈ 4.6 min) without stalling?

- **How Our Findings & Mechanistic Model Close This Gap:**

Our two-site kinetic parameters and competition hierarchy integrate the vascular laminin literature [1–5] with the MCAM ectodomain-shedding literature [10–12] into a unified three-step physical mechanism:

1. **Isoform-Selective Pathfinding ( `MCAM` Selects α₄ Over α₅ Autonomous of the β₁γ₁ Coiled-Coil):** Unlike α₆β₁ integrin—which strictly requires the intact αβγ coiled-coil rod and C-terminal γ₁-E1605 tail to bind any laminin and exhibits zero binding to isolated α-chain `LG1–LG3` [7, 8]—MCAM directly binds isolated Laminin-α₄ `LG1–LG3` ( `LamA4G123` , lacking β₁γ₁ chains; K_D,1:1 = 584 ± 79 nM, K_D,int = 824 ± 36 nM, N = 3). In intact heterotrimeric Laminin-411 ( `tE411` ), pre-organization of `LG1–LG2` by the α₄β₁γ₁ coiled-coil tightens MCAM's initial single-site encounter **~ 24-fold** (K_D,int = 34.4 ± 2.3 nM, k,on,int = 1.13 × $10^6$ $M^{-1}s^{-1}$) and drives dominant bidentate clamping (K_D,full = 4.63 ± 0.48 nM, $\tau_1/_2$ ≈ 4.6 min) selectively on Laminin-411/421 permissive portals.

2. **Dominant α₆β₁ Integrin Traction on Adjacent Laminin-411 Molecules:** Once the cell is positioned on a Laminin-411–rich zone in 2D ( `Panel A` ), cell-surface α₆β₁ integrins (K_D = 0.73 ± 0.22 nM, >6× tighter than

MCAM) engage neighboring Laminin-411 molecules in the basement membrane network to transmit actomyosin traction forces.

3. **Juxtamembrane Proteolytic Shedding Releases the Trailing MCAM Tether:** Because the two-site MCAM–Laminin-411 latch dissociates slowly ($k_{off,full} = 2.53 \times 10^{-3}\ s^{-1}$) and $\alpha_6\beta_1$ integrin does not allosterically accelerate MCAM dissociation, **proteolytic cleavage of juxtamembrane MCAM by cell-surface metalloproteinases (e.g., MT1-MMP/MMP-14)** [10–12] provides an irreversible release mechanism during migration. Crucially, shedding detaches the cell while **leaving the cleaved `sMCAM` ectodomain bound as a molecular footprint on that individual Laminin-411 molecule**, without disturbing $\alpha_6\beta_1$ integrins docked on adjacent Laminin-411 molecules.

| Construct / Surface-Binding Model Parameter | Chains Present | Apparent K_D (Mean ± SD, N=3) | k,on ($M^{-1}s^{-1}$) | k,off ($s^{-1}$) | Dissociation $t_{1/2}$ | Component / Step Contribution | Global $R^2$ |
|---|---|---|---|---|---|---|---|
| **Intact Laminin-411 ( `tE411` ) — Apparent Step 1 (K_D,step 1, Sequential Capture)** | $\alpha_4\beta_1\gamma_1$ (with coiled-coil) | **4.63 ± 0.48 nM** | (5.44 ± 0.25) × $10^5$ | (2.53 ± 0.36) × $10^{-3}$ | **278 ± 41 s** | **58.9–89.3%** (slow-off step 1) | 0.9970 |
| **Intact Laminin-411 ( `tE411` ) — Apparent Step 2 (K_D,step 2, Sequential Capture)** | $\alpha_4\beta_1\gamma_1$ (with coiled-coil) | **34.4 ± 2.3 nM** | (1.13 ± 0.05) × $10^6$ | (3.88 ± 0.41) × $10^{-2}$ | **18.0 ± 1.9 s** | 10.7–41.1% (fast-off step 2) | 0.9970 |
| **Intact Laminin-411 ( `tE411` ) — Global 1:1 Fit ( `0–240 s` )** | $\alpha_4\beta_1\gamma_1$ (with coiled-coil) | **17.6 ± 0.4 nM** | (4.17 ± 0.22) × $10^5$ | (7.34 ± 0.47) × $10^{-3}$ | **94.7 ± 6.2 s** | 100% (single-site approx.) | 0.9802 |
| **Isolated `LamA4G123` — Apparent Pop 2 (K_D,pop 2, Two-Population 2:1)** | $\alpha_4$ `LG1–LG3` only (no βγ) | **824 ± 36 nM** | (1.37 ± 0.16) × $10^5$ | (1.13 ± 0.13) × $10^{-1}$ | **6.2 ± 0.7 s** | **86.1 ± 1.0%** (dominant fast) | 0.9987 |
| **Isolated `LamA4G123` — Apparent Pop 1 (K_D,pop 1, Two-Population 2:1)** | $\alpha_4$ `LG1–LG3` only (no βγ) | **25.4 ± 18.1 nM** | (8.66 ± 1.64) × $10^4$ | (2.25 ± 1.69) × $10^{-3}$ | **439 ± 278 s** | **13.9 ± 1.0%** (minor slow tail) | 0.9987 |
| **Isolated `LamA4G123` — Global 1:1 Fit ( `Fit` , `0–135 s` )** | $\alpha_4$ `LG1–LG3` only (no βγ) | **584 ± 79 nM** | (1.13 ± 0.05) × $10^5$ | (6.62 ± 1.14) × $10^{-2}$ | **10.7 ± 2.0 s** | 100% (single-site approx.) | 0.9973 |
| **$\alpha_6\beta_1$ Integrin + Intact Laminin-411 (Competition)** | $\alpha_4\beta_1\gamma_1$ (with coiled-coil) | **0.73 ± 0.22 nM** | — | — | — | 98.5 ± 1.4% block at 40 nM | 0.9818 |
| **$\alpha_6\beta_1$ Integrin + Isolated α-Chain `LG1–LG3`** | α `LG1–LG3` only (no βγ) | **No Binding** | — | — | — | 0% (requires $\beta_1\gamma_1$ & $\gamma_1$-E1605) | Takizawa *et al.*, 2017 |

**Table S1. Quantitative Comparison: Intact Laminin-411 (α4β1γ1) vs. Isolated LamA4G123 (α4 LG1–LG3 Only)**

*KD,1 = Combined High-Affinity State (Both Sites Engaged: β1γ1 + α4 LG1–LG3); KD,2 = Low-Affinity State (1 Site Bound) across N = 3 BLI replicates.*

| Construct / Surface-Binding Model Parameter | Chains Present | Apparent KD (Mean ± SD, N = 3) | kon ($M^{-1}s^{-1}$) | koff ($s^{-1}$) | Dissociation $t_{1/2}$ | Component / Step Contribution | Global $R^2$ |
|---|---|---|---|---|---|---|---|
| **Intact Laminin-411 (tE411) — Both Sites Engaged (KD,1 / KD,step1)** | α4β1γ1 (with coiled-coil) | **4.63 ± 0.48 nM** | (5.44 ± 0.25) × $10^5$ | (2.53 ± 0.36) × $10^{-3}$ | **278 ± 41 s** | 58.9–89.3% (both sites bound) | 0.9970 |
| **Intact Laminin-411 (tE411) — 1 Site Bound (KD,2 / KD,step2)** | α4β1γ1 (with coiled-coil) | **34.4 ± 2.3 nM** | (1.13 ± 0.05) × $10^6$ | (3.88 ± 0.41) × $10^{-2}$ | **18.0 ± 1.9 s** | 10.7–41.1% (1 site bound) | 0.9970 |
| **Intact Laminin-411 (tE411) — Global 1:1 Fit (0-240 s)** | α4β1γ1 (with coiled-coil) | **17.6 ± 0.4 nM** | (4.17 ± 0.22) × $10^5$ | (7.34 ± 0.47) × $10^{-3}$ | **94.7 ± 6.2 s** | 100% (single-site approx.) | 0.9802 |
| **Isolated LamA4G123 — 1 Site Bound (KD,2 / KD,pop2, α4 LG1–LG3 Only)** | α4 LG1–LG3 only (no βγ) | **824 ± 36 nM** | (1.37 ± 0.16) × $10^5$ | (1.13 ± 0.13) × $10^{-1}$ | **6.2 ± 0.7 s** | 86.1 ± 1.0% (1 site bound) | 0.9987 |
| **Isolated LamA4G123 — Residual Slow Component (KD,1 / KD,pop1)** | α4 LG1–LG3 only (no βγ) | **25.4 ± 18.1 nM** | (8.66 ± 1.64) × $10^4$ | (2.25 ± 1.69) × $10^{-3}$ | **439 ± 278 s** | 13.9 ± 1.0% (minor slow tail) | 0.9987 |
| **Isolated LamA4G123 — Global 1:1 Fit (Fit, 0-135 s)** | α4 LG1–LG3 only (no βγ) | **584 ± 79 nM** | (1.13 ± 0.05) × $10^5$ | (6.62 ± 1.14) × $10^{-2}$ | **10.7 ± 2.0 s** | 100% (single-site approx.) | 0.9973 |
| **α6β1 Integrin + Intact Laminin-411 (Competition)** | α4β1γ1 (with coiled-coil) | **0.73 ± 0.22 nM** | — (Equilibrium SET) | — | **—** | 98.5 ± 1.4% block at 40 nM | 0.9818 |
| **α6β1 Integrin + Isolated α-Chain LG1-LG3** | α LG1–LG3 only (no βγ) | **No Binding** | — | — | **—** | 0% (requires β1γ1 & γ1-E1605) | Takizawa 2017 |

Values represent Mean ± SD across N = 3 independent replicates. MCAM-Fc is immobilized on Protein A sensors via its Fc tag (no solution Fc avidity); KD,1 = 4.63 ± 0.48 nM is the combined KD when both sites on Laminin-411 (β1γ1 + α4 LG1–LG3) are engaged (parameterized as apparent stepwise/component constants KD,step1, KD,step2 and KD,pop1, KD,pop2).

*Supplementary Table 1 — Quantitative Kinetic & Equilibrium Summary Comparing Intact Heterotrimeric Laminin-411 (tE411) vs. Isolated Laminin-alpha4 LG1-LG3 (LamA4G123)*

**Table S2. Solution Competition / Inhibition of Laminin-411 (20 nM) Binding to MCAM-Fc by Soluble α6β1 Integrin (N = 3)**

*Top rows: Quadratic solution KD under linear plateau, saturable detector (R = Rmax a / (KD,capture + a)), and initial-rate (v0, Day et al. 2013) calibrations; model half-depletion point ([I]50,quad) vs. empirical Hill IC50.*

| Solution Competition Parameter / [α6β1 Integrin] Dose | Symbol / Molar Ratio | Derived Free [LM411] (% Bound) | Mean ± SD (N = 3) | SEM | Rep 1 | Rep 2 | Rep 3 | 1:1 Mass-Action Fit / $R^2$ |
|---|---|---|---|---|---|---|---|---|
| **Quadratic Solution Affinity — Linear Plateau Mapping (R ∝ [LM411]free)** | KD(α6β1)lin [nM] | a50 = 10.00 nM (50.0% bound) | **0.73 ± 0.22 nM** | 0.13 nM | 0.98 nM | 0.54 nM | 0.66 nM | $R^2$ = 0.9818 |
| **Quadratic Solution Affinity — Conditional on Saturable Detector (KD,capture = 34.4 nM)** | KD(α6β1)cal [nM] | a50,resp = 7.75 nM (61.3% bound) | **0.26 ± 0.08 nM** | 0.05 nM | 0.35 nM | 0.21 nM | 0.24 nM | $R^2$ = 0.9900 |
| **Quadratic Solution Affinity — Initial Rate (v0 = dR/dt, Day et al. 2013)** | KD(α6β1)v0 [nM] | a50 = 10.00 nM (50.0% bound) | **0.55 ± 0.40 nM** | 0.23 nM | 0.97 nM | 0.17 nM | 0.52 nM | $R^2$ = 0.9867 |
| **Quadratic Model Half-Depletion Point ([I]50,quad = ½[LM411]tot + KD,lin)** | [I]50,quad [nM] | 10.00 nM (50.0% free LM411) | **10.73 ± 0.22 nM** | 0.13 nM | 10.98 nM | 10.54 nM | 10.66 nM | Cal: 10.26 ± 0.08 nM |
| **Calibrated 50%-Response Midpoint (R = 0.5 R0 at a50,resp = 7.75 nM)** | IC50,resp(cal) [nM] | 7.75 nM (61.3% bound LM411) | **12.67 ± 0.12 nM** | 0.07 nM | 12.80 nM | 12.58 nM | 12.62 nM | $R^2$ = 0.9900 |
| **Empirical 4PL/Hill Fit Midpoint (Separate Empirical Fit, h = 2.66 ± 0.25)** | IC50,Hill [nM] | Empirical 50% BLI Response | **11.53 ± 0.54 nM** | 0.31 nM | 12.06 nM | 10.98 nM | 11.55 nM | $R^2$Hill = 0.9964 |
| **Relative Apparent Affinity Ratio (KD,step1(MCAM) / KD(α6β1)lin)** | KD,step1(MCAM) / KD(α6β1) | MCAM = 4.63 nM | **6.36 ± 2.40×** | 1.39× | 4.18× | 8.98× | 7.61× | Cal: 17.8 ± 5.5× |
| **0.00 nM α6β1 Integrin (Uninhibited 20 nM LM411 Control)** | Ratio = 0.0000 | 20.00 nM (0.0% bound) | **100.00 ± 0.00%** | 0.00% | 100.00% | 100.00% | 100.00% | 100.00% (0.311 nm) |
| **1.25 nM α6β1 Integrin** | Ratio = 0.0625 | 18.80 nM (6.0% bound) | **99.91 ± 1.33%** | 0.77% | 101.13% | 98.50% | 100.09% | 93.98% (0.311 nm) |
| **2.50 nM α6β1 Integrin** | Ratio = 0.1250 | 17.60 nM (12.0% bound) | **101.54 ± 1.65%** | 0.95% | 101.96% | 99.72% | 102.95% | 87.99% (0.316 nm) |
| **5.00 nM α6β1 Integrin** | Ratio = 0.2500 | 15.23 nM (23.9% bound) | **85.33 ± 2.20%** | 1.27% | 86.29% | 82.81% | 86.89% | 76.12% (0.266 nm) |
| **10.00 nM α6β1 Integrin (0.5× Sub-Stoichiometric)** | Ratio = 0.5000 | 10.64 nM (46.8% bound) | **62.22 ± 3.38%** | 1.95% | 65.43% | 58.70% | 62.53% | 53.16% (0.194 nm) |
| **20.00 nM α6β1 Integrin (1:1 Equimolar Ratio)** | Ratio = 1.0000 | 3.47 nM (82.6% bound) | **17.05 ± 1.74%** | 1.01% | 18.99% | 15.64% | 16.50% | 17.22% (0.053 nm) |
| **40.00 nM α6β1 Integrin (2:1 Molar Excess → 98.5% Block)** | Ratio = 2.0000 | 0.68 nM (96.6% bound) | **1.53 ± 1.44%** | 0.83% | 3.12% | 0.30% | 1.16% | 3.35% (0.005 nm) |

Response calibration: Under linear mapping (R ∝ a = [LM411]free), KD = 0.73 ± 0.22 nM and the model-derived half-depletion point (a = 10.0 nM) is [I]50,quad = ½[LM411]tot + KD = 10.73 ± 0.22 nM (separate empirical 4PL/Hill fit: IC50,Hill = 11.53 ± 0.54 nM).
Under the saturable-detector model R(a) = Rmax a / (KD,capture + a) (KD,capture = 34.4 nM), halving the BLI response occurs at a50,resp = 7.75 nM free LM411 (IC50,resp = 12.67 ± 0.12 nM; KD,cal = 0.26 ± 0.08 nM, conditional on the specified detector model unless KD,capture has an independent calibration).

*Supplementary Table 2 — Solution Equilibrium Competition Parameters for alpha6beta1 Integrin vs. Immobilized MCAM-Fc on Intact Laminin-411 (20 nM tE411, 0-40 nM Integrin, N = 3)*

**Table S3. Direct Binding Kinetics of Intact Laminin-411 (α4β1γ1 Trimer) to MCAM-Fc — Per-Curve & Apparent Stepwise Parameters**

*Global Apparent Stepwise Parameters (N = 3): KD,step1 = 4.63 ± 0.48 nM (kon1 = 5.44×$10^5$ $M^{-1}s^{-1}$, koff1 = 2.53×$10^{-3}$ $s^{-1}$, t1/2 = 278 s) | KD,step2 = 34.4 ± 2.3 nM (kon2 = 1.13×$10^6$ $M^{-1}s^{-1}$, koff2 = 3.88×$10^{-2}$ $s^{-1}$, t1/2 = 18.0 s)*

| [Laminin-411] (nM) | Rmax,step1 (nm) | Rmax,step2 (nm) | Rmax,tot (nm) | Step 1 (%) | kobs,1 ($s^{-1}$) [$t_{1/2}$,on] | kobs,2 ($s^{-1}$) [$t_{1/2}$,on] | Obs. vs. Fit R(120 s) (nm) | Obs. vs. Fit R(240 s) (nm) | Retained at 240 s (%) | Curve $R^2$ (RMSE) |
|---|---|---|---|---|---|---|---|---|---|---|
| **125.00 nM** | 0.611 ± 0.006 | 0.426 ± 0.003 | 1.037 ± 0.005 | **58.9 ± 0.4%** | 0.0706 ± 0.0035 [9.8 s] | 0.1797 ± 0.0097 [3.9 s] | 0.957 ± 0.012 (0.965) | 0.414 ± 0.026 (0.426) | 43.2 ± 2.2% | **0.9938 (0.0182 nm)** |
| **62.50 nM** | 0.591 ± 0.009 | 0.315 ± 0.007 | 0.906 ± 0.012 | **65.2 ± 0.6%** | 0.0366 ± 0.0019 [19.0 s] | 0.1093 ± 0.0069 [6.4 s] | 0.804 ± 0.018 (0.808) | 0.362 ± 0.020 (0.384) | 45.1 ± 1.4% | **0.9953 (0.0130 nm)** |
| **31.25 nM** | 0.595 ± 0.014 | 0.153 ± 0.011 | 0.748 ± 0.018 | **79.6 ± 1.2%** | 0.0195 ± 0.0011 [35.5 s] | 0.0741 ± 0.0055 [9.4 s] | 0.599 ± 0.019 (0.600) | 0.310 ± 0.019 (0.317) | 51.8 ± 1.5% | **0.9893 (0.0142 nm)** |
| **15.63 nM** | 0.494 ± 0.036 | 0.065 ± 0.014 | 0.560 ± 0.050 | **88.4 ± 1.4%** | 0.0110 ± 0.0007 [63.0 s] | 0.0564 ± 0.0048 [12.3 s] | 0.316 ± 0.026 (0.326) | 0.196 ± 0.019 (0.193) | 61.9 ± 1.0% | **0.9902 (0.0074 nm)** |
| **7.81 nM** | 0.394 ± 0.041 | 0.048 ± 0.013 | 0.441 ± 0.053 | **89.3 ± 1.8%** | 0.0068 ± 0.0005 [102.6 s] | 0.0476 ± 0.0044 [14.6 s] | 0.147 ± 0.014 (0.153) | 0.103 ± 0.010 (0.099) | 69.9 ± 0.8% | **0.9867 (0.0040 nm)** |

Derived observed association rate: kobs = kon[C] + koff; association half-time: t1/2,on = ln 2 / kobs. Retained at 240 s (%) = 100 × R(240 s) / R(120 s).

*Supplementary Table 3 — Per-Curve & Replicate Fitted Parameters for Intact Laminin-411 (tE411, 7.81-125 nM, N = 3) Binding to Immobilized MCAM-Fc*

**Table S4. Direct Binding Kinetics of Isolated LamA4G123 (α4 LG1-LG3 Without βγ Chains) to MCAM-Fc — Per-Curve & Apparent Component Parameters**

*Global Two-Population (2:1) Parameters (N = 3): Apparent Pop 2 KD,pop2 = 824 ± 36 nM (kon2 = 1.37×10⁵ $M^{-1}s^{-1}$, koff2 = 0.113 $s^{-1}$, t1/2 = 6.2 s, ~86%) | Apparent Pop 1 KD,pop1 = 25.4 ± 18.1 nM (~14%) | Global 1:1 KD = 584 ± 79 nM*

| [LamA4G123] (nM) | Rmax,pop1 (nm) | Rmax,pop2 (nm) | Rmax,tot (nm) | Population 1 (%) | kobs,1 ($s^{-1}$) [$t_{1/2}$,on] | kobs,2 ($s^{-1}$) [$t_{1/2}$,on] | Obs. vs. Fit R(120 s) (nm) | Obs. vs. Fit R(240 s) (nm) | Retained at 240 s (%) | Curve $R^2$ (RMSE) |
|---|---|---|---|---|---|---|---|---|---|---|
| **500.00 nM** | 0.088 ± 0.015 | 0.543 ± 0.054 | 0.631 ± 0.069 | **13.9 ± 1.0%** | 0.0455 ± 0.0090 [15.7 s] | 0.1819 ± 0.0202 [3.8 s] | 0.294 ± 0.031 (0.289) | 0.067 ± 0.008 (0.064) | 22.9 ± 3.7% | **0.9982 (0.0042 nm)** |
| **250.00 nM** | 0.060 ± 0.049 | 0.522 ± 0.046 | 0.581 ± 0.087 | **9.7 ± 6.6%** | 0.0239 ± 0.0050 [30.0 s] | 0.1475 ± 0.0164 [4.7 s] | 0.172 ± 0.049 (0.171) | 0.037 ± 0.023 (0.035) | 19.9 ± 6.9% | **0.9979 (0.0026 nm)** |
| **125.00 nM** | 0.037 ± 0.007 | 0.468 ± 0.094 | 0.506 ± 0.100 | **7.4 ± 0.9%** | 0.0131 ± 0.0031 [55.4 s] | 0.1303 ± 0.0146 [5.4 s] | 0.085 ± 0.017 (0.086) | 0.025 ± 0.002 (0.018) | 30.2 ± 7.6% | **0.9960 (0.0018 nm)** |
| **62.50 nM** | 0.033 ± 0.011 | 0.391 ± 0.121 | 0.424 ± 0.128 | **7.8 ± 2.4%** | 0.0077 ± 0.0023 [96.7 s] | 0.1217 ± 0.0137 [5.7 s] | 0.041 ± 0.012 (0.041) | 0.014 ± 0.002 (0.010) | 36.4 ± 5.7% | **0.9884 (0.0013 nm)** |
| **31.25 nM** | 0.015 ± 0.014 | 0.375 ± 0.163 | 0.390 ± 0.176 | **3.3 ± 2.0%** | 0.0050 ± 0.0019 [155.9 s] | 0.1174 ± 0.0132 [6.0 s] | 0.019 ± 0.011 (0.017) | 0.004 ± 0.007 (0.003) | 9.1 ± 43.2% | **0.9747 (0.0010 nm)** |
| **15.63 nM** | 0.056 ± 0.015 | 0.225 ± 0.234 | 0.281 ± 0.237 | **42.6 ± 49.7%** | 0.0036 ± 0.0018 [227.5 s] | 0.1153 ± 0.0130 [6.1 s] | 0.010 ± 0.006 (0.011) | 0.009 ± 0.002 (0.005) | 102.5 ± 35.6% | **0.6838 (0.0011 nm)** |

7.81 nM trace omitted as sub-threshold baseline noise. Global 1:1 fit over the 0–135 s Gator window yields KD,1:1 = 583.7 ± 79.4 nM ($R^2$ = 0.9973).

*Supplementary Table 4 — Per-Curve & Replicate Fitted Parameters for Isolated Laminin-alpha4 LG1-LG3 (LamA4G123, 15.63-500 nM, N = 3) Binding to Immobilized MCAM-Fc*

### Gap 4: How 3D Soluble `sMCAM` (`sCD146`) Acts as a Pericellular Brake & How the Structure Guides Epitope-Selective Biologics

- **What Was Outstanding:**

Boneberg et al. (2009) [10] and subsequent studies [11, 12] established that $Ca^{2+}$ influx and metalloproteinase activity trigger robust ectodomain shedding of soluble `sMCAM` (`sCD146`) from endothelial and tumor cells. Yet whether shed `sMCAM` has a direct mechanical role on the extracellular matrix—and why micromolar concentrations (1 µM) of monomeric `sMCAM` are needed in vitro to block cell migration—remained unexplained. Moreover, while Patarroyo and colleagues showed that certain anti-Laminin-$\alpha_4$ antibodies inhibit migration [6], the precise structural target within the ~1,000-residue `LG1–LG5` tandem was unknown.

- **How Our Data Close This Gap:**

  1. **3D Matrix Pre-Saturation (`Panel B` of Conclusion Model):** When soluble `sMCAM` is present at high concentration in **3D solution** (either via extensive pericellular shedding or exogenous addition), it is no longer confined to the 2D cell footprint. It diffuses freely in 3D to **pre-occupy `LG1–LG2` across the unengaged Laminin-411 matrix ahead of the cell**, sterically blocking $\alpha_6\beta_1$ integrin docking and arresting migration.

  2. **Quantitative Explanation for Competitor Dosing (1 µM):** By Cheng–Prusoff mass action ($IC_{50}$ = K_D,MCAM(1 + [Integrin],eff / K_D,Integrin)), a soluble decoy with K_D,full = 4.63 nM (K_D,int = 34.4 nM) competing against a **0.72 nM integrin receptor** reinforced by multivalent 2D cell-surface avidity requires 0.2–1 µM to maintain >95% fractional saturation of matrix `LG1–LG2`.

  3. **Atomic Blueprint for Epitope-Selective Anti-LAMA4 (`LG1–LG2`) Biologics:** Systemic blockade of $\alpha_6\beta_1$ integrin causes severe epithelial toxicity because $\alpha_6\beta_1$ mediates essential adhesion to epithelial Laminin-111 and Laminin-511 [7–9]. By pinpointing the shared MCAM/$\alpha_6\beta_1$ interface specifically to **Laminin-$\alpha_4$ `LG1–LG2` (sparing `LG3` – `LG5`)**, our cryo-EM structures explain why only a subset of anti-$\alpha_4$ `LG` antibodies block migration [6] and provide the atomic template to engineer epitope-selective anti-LAMA4 `LG1–LG2` antibodies or bivalent `MCAM–Fc` biologics.

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## 3. Summary Table: What Was Outstanding vs. How This Study Resolves It

| Mechanistic Question / Paradox | Prior State of the Field | Resolution Provided by Our Study ( `PMC13596217` + MIXA Biophysics) |
|---|---|---|
| **1. How does MCAM recognize Laminin-411/421, and does it overlap the $\alpha_6\beta_1$ integrin site?** | Unknown which of the five $\alpha_4$ `LG1–LG5` domains bind MCAM or whether overlap with $\alpha_6\beta_1$ is direct or allosteric [3, 5–8]. | **Cryo-EM + Mutagenesis + Off-Rate Invariance:** MCAM `V2–C1` clamps directly across Laminin-$\alpha_4$ **`LG1–LG2`** ( `LG3` uninvolved), directly overlapping the $\alpha_6\beta_1$ footprint via pure orthosteric competition (k,off invariant, p = 0.41). |
| **2. Why doesn't co-expressed membrane MCAM block $\alpha_6\beta_1$ integrin migration in 2D?** | Puzzling paradox: TH17 and melanoma cells co-express both receptors yet migrate rapidly via $\alpha_6\beta_1$ integrin [3, 5]. | **Quantitative Affinity Hierarchy + 2D Restriction ( `Panel A` ):** $\alpha_6\beta_1$ integrin binds `LG1–LG2` with **K_D = 0.72 ± 0.14 nM—6.4× tighter** than 2-site MCAM (4.63 ± 0.48 nM) and **48× tighter** than 1-site MCAM (34.4 ± 2.3 nM). Confined in 2D, membrane MCAM cannot outcompete $\alpha_6\beta_1$. |
| **3. How do cells select $\alpha_4$ exit portals and release MCAM during diapedesis?** | $\alpha_6\beta_1$ binds inhibitory Laminin-511 ($\alpha_5$) tighter than Laminin-411 ($\alpha_4$) [1, 2, 9]; mechanism of MCAM release was unclear. | **Two-Gear Capture + Shedding Release:** MCAM selectively captures $\alpha_4$ (K_D,int = 34.4 nM → K_D,full = 4.63 nM, $\tau_{1/2} \approx$ 4.6 min); $\alpha_6\beta_1$ (0.72 nM) pulls on adjacent LM411; MMP shedding [10] cleaves trailing MCAM, leaving `sMCAM` on its own LM411 molecule. |
| **4. Why does soluble `sMCAM` require ~ 1 µM to arrest migration, and how should $\alpha_4$ be targeted?** | `sMCAM` (1 µM) was assumed by Reviewer #1 to have "very low affinity"; anti-$\alpha_4$ mAbs had unmapped epitopes [6]. | **3D Pre-Saturation ( `Panel B` ) & Epitope Blueprint:** High 3D `sMCAM` globally saturates matrix `LG1–LG2` ahead of the cell; 1 µM is required by Cheng–Prusoff mass action to block a 0.72 nM avidity-enhanced integrin. Defines `LG1–LG2` as the exact epitope for selective biologics. |

## 4. Full Numbered Bibliography (Citations)


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